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Optimal Policy for Financial Market Tokenization

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Optimal Policy for Financial Market Tokenization
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ABSTRACT: Most tokenized financial assets are on restricted-access ("private") blockchains and represent underlying assets that brokers deposit at custodians. Trading with tokens unlocks efficiency gains, but unconnected private blockchains also risk fragmenting markets, leading policymakers in several countries to explore interoperability mandates or public-private tokenization partnerships. We provide the first formal model of tokenized market formation and use it to derive optimal policy. Brokers with heterogeneous market power compete to attract investors and execute their trades, which is cheaper within a tokenized market that a coalition of brokers can set up. Partial coalitions divert trades away from excluded competitors, leading to equilibrium coalition structures that can feature excessive investment or insufficient tokenization. Indeed, technological progress that facilitates asset tokenization can increase fragmentation. Neither public-private cost-sharing nor interoperability mandates can guarantee the social optimum when used alone, but their optimal combination does.

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1 Introduction

The financial press is abuzz with initiatives by brokers to tokenize financial assets.[1] Tokenization entails representing an asset on a shared ledger using a token that can embed self-executing contracts (Agur et al., 2026; Aldasoro et al., 2023). This programmability in turn enables contingent transactions that could reduce trading costs. For instance, a tokenized ledger enables the simultaneous and instantaneous exchange of a seller's asset and a buyer's payment, reducing the need for costly intermediaries like clearinghouses and registrars. J.P. Morgan (2023) estimates that the continuous reinvestment of cash that is currently frozen during settlement could reduce portfolio management costs by 22%, while estimates on matched samples of tokenized and conventional bonds find that tokenization lowers yield spreads by 24-40% (Aldasoro et al., 2025; Born et al., 2026; Leung et al., 2023).[2]

Most financial asset tokenization to date has occurred on private ledgers (e.g., Broadridge, Hadron, Kinexys), rather than on public blockchains such as Ethereum (Figure 1). Private ledgers limit access to a (potentially small) set of permissioned institutions and may not be interoperable with each other. The proliferation of such ledgers thus raises concerns that asset trade will fragment (e.g., BIS, 2026; IMF, 2026) and policy institutions are responding. Current approaches range from ensuring the interoperability of private ledgers (as in Hong Kong, Singapore, Thailand, and the UK) to public-private partnerships where the public sector directly supports the development of the new market infrastructure (as considered in Australia, Brazil, the Eurozone, India, and the multinational Project Agorá).[3]

What economic forces drive brokers toward "fragmented" tokenization and why is such fragmentation socially costly? What trade-offs should policymakers consider when determining whether and how to intervene? Does requiring interoperability suffice, or should the public sector invest directly in forming tokenized markets? To help answer these questions, in this paper we develop the first formal analysis of the formation of tokenized markets.

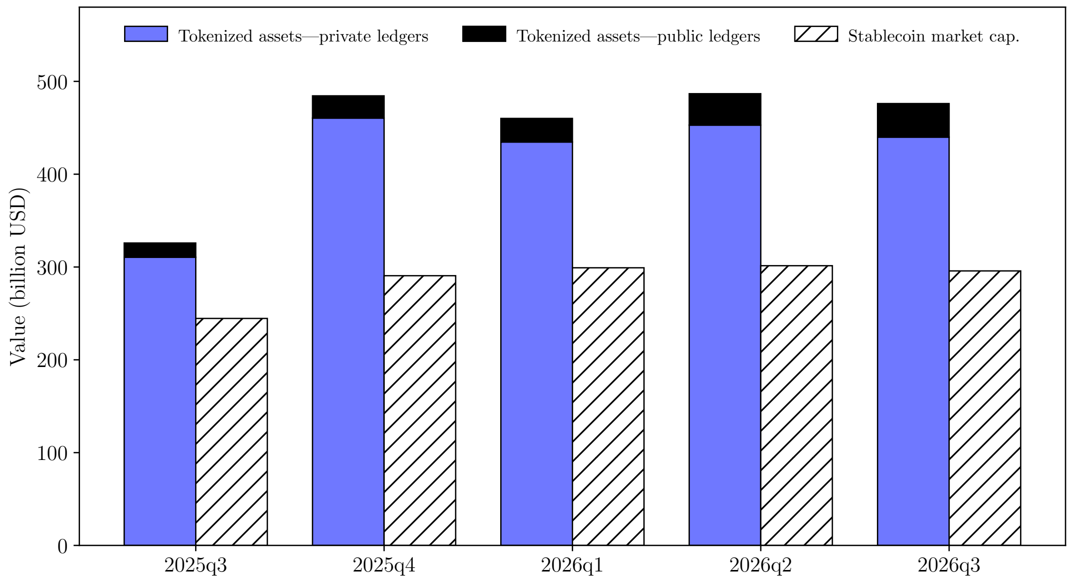
We model an island economy containing investors who trade with one another through brokers. Investors are born with a money endowment and one asset and choose a broker at which to deposit their asset. This makes them a client of the broker, which has value because at a future date a preference shock (such as from liquidity needs) makes some investors want to sell their asset while others take the buy side.

[1] See, for instance, recent initiatives by BlackRock, Goldman Sachs and J.P. Morgan.

[2] For estimates of the efficiency gains from tokenization for various categories of financial assets, see also: Agur et al. (2026); Budish and Sunderam (2026); Cong et al. (2025a); DFCRC (2026); Harvey et al. (2026); IMF (2026); Lin et al. (2026); Liu, Shim and Zheng (2023).

[3] In the US, policymakers are permitting two separate tokenized systems to emerge, with interoperability within but not between systems. Public support can also include providing tokenized central-bank money or government bonds; we discuss such initiatives in Section 7 below.

Figure 1: Tokenization by brokers on private and public ledgers



Notes: The stacked blue and black bars show the value of tokenized assets on, respectively, private and public ledgers; the hatched bars show stablecoin market capitalization. Values are at quarter starts. We measure these quantities using, respectively, the "represented," "distributed," and stablecoin categories on rwa.xyz, which defines *represented* tokenized assets as "using the blockchain as a recordkeeping layer [...] without enabling onchain investor transfer or distribution," whereas *distributed* tokenized assets use "the blockchain as a distribution layer, enabling onchain investors to subscribe, hold and manage assets directly." The data on rwa.xyz include tokenized bonds, commodities, equity, private credit, and repurchase agreements and, to our knowledge, fully cover tokenization on public but not private ledgers—data on some known private ledgers like Hadron and Kinexys (with over 7 billion US dollars of daily transactions) is lacking. Thus, the figure may understate the size of tokenization on private ledgers.

Each island contains only one broker and a number of investors. Investors incur a distance cost if they opt for an off-island broker, giving each broker a degree of market power. Our baseline model consists of three islands containing, respectively, one, two and three local investors. This is the simplest structure that delivers two key features: heterogeneity in broker market power and the possibility of fragmentation of inter-broker trade. The importance of the former is well-established in the literature (Duffie, 2022; Dugast, Üslü and Weill, 2022) while the latter is our focus and requires at least three brokers.

Brokers begin with only a costly "legacy" settlement technology, but can choose whether to participate in a shared, costly investment to form a tokenized market. Brokers choosing to tokenize deliver the assets that they receive from investors to a custodian, then can trade tokens representing the custodied assets with one another. Settlement on the tokenized market is costless. However, because tokenization locks the underlying asset at a custodian, it creates a new friction when settling trades with brokers that only use the legacy technology. Specifically, trading a tokenized asset with a legacy broker involves an extra cost, reflecting the processing time and cost involved in destroying the token and obtaining the release of the asset from the custodian.[4]

In addition to inter-broker trade, each broker can also process trades internally among its own clients, if there are feasible matches. Such intra-broker trade is costless to settle—reflecting, for instance, that within a broker there is no settlement risk.

Facing these trading costs, each broker chooses to sequence its feasible trades as follows: first, it settles the available intra-broker trades; second, it settles with other brokers that use the same ledger technology; third, if any feasible trades remain, it settles with brokers using the opposite technology.

Brokers earn revenue from transaction fees and can price discriminate between investors based on their location. On each island, brokers then engage in Bertrand competition subject to differentiation on the expected costs of settling transactions and the matching probabilities they offer investors, both of which depend on trading technology choices.[5]

At the start of the game, brokers negotiate about forming a tokenized market. Weighing the implications for their expected profits, the brokers engage in a coalition formation game that results in no coalition, a partial coalition of any two brokers, or the grand coalition of three brokers. An equilibrium is found when none wish to individually or jointly deviate to a different coalition.

We find that in equilibrium the grand coalition never forms. When brokers find a tokenized market too costly to set up, no coalition forms. In contrast, when brokers find that the benefits of a tokenized market justify its cost, a partial coalition forms. The underlying force is trade diversion. When one broker is excluded from the tokenized market, the included brokers expect to process a larger share of all investor matches and can also gain market power, allowing them to extract higher fees from their local investors. This outweighs the desire to minimize the cost of the average trade, which is zero when the legacy market is retired under the grand coalition. In equilibrium, the largest broker (Broker 3) and the smallest (Broker 1) band together to divert trade from the third (Broker 2).

A partial coalition is socially suboptimal. Welfare—defined as the sum of investor utility and broker profit—is highest under either the grand coalition or no coalition, depending on

[4] For details on the costs involved in tokenizing and de-tokenizing assets, see, for instance: DTCC (2026); Patel et al. (2026); SEC (2025, 2026).

[5] All investors looking to take a given side of the transaction place the same value on the potential trade. Investors' heterogeneous access to trading technologies via brokers then determines their likelihood of a match. We thus abstract from heterogeneous asset valuations and bidding for tractability, centering our model instead on the role of endogenous market infrastructure.

the setup cost and efficiency gains of the tokenized market. Intuitively, if the setup cost of the tokenized market is to be incurred, all brokers should be included on this market to eliminate costly legacy market transactions. When partial coalitions form in equilibrium, they instead involve either excessive investment or insufficient tokenization. Excessive investment occurs if a tokenized market socially costs more than it is worth, but the private benefit of trade diversion nonetheless leads to its formation through a partial coalition. Insufficient tokenization results when forming the tokenized market maximizes welfare, but trade diversion incentives lead two brokers to exclude the third from the market.

Interestingly, reducing the friction involved in settling trades that span tokenized and legacy markets can actually *reduce* welfare in equilibrium. The token-to-legacy settlement cost disincentivizes partial coalitions, so when this friction declines brokers become more inclined to form such coalitions, expanding the set of parameter values for which excessive investment occurs. Thus, our model highlights that technological or regulatory developments that facilitate the quick release of underlying assets from custodians (see, e.g., BIS, 2024; IOSCO, 2025) could exacerbate market fragmentation, if pursued in isolation.

Turning to policy, we first consider an interoperability mandate, which gives an initially excluded broker the option to subsequently join the tokenized market for free. We find that while interoperability removes trade diversion incentives, it does not guarantee the social optimum—instead, it can produce an underinvestment problem. This stems from two sources. First, part of the gains from the grand coalition flow to investors, due to the increased competition among brokers, which lowers the fees that the largest broker can charge. Second, gains from forming the grand coalition are not spread evenly across brokers. Both mechanisms imply that the preferences of the marginal broker, whose choice determines whether the grand coalition forms, do not align with those of a social planner.

Escaping the underinvestment problem requires a second instrument: subsidizing tokenized market formation through lump-sum taxes on investors, in the vein of public-private tokenization initiatives. When used alone, this tool never improves welfare. However, the combination of a tokenization subsidy and an interoperability mandate *can* achieve the socially optimal outcome in all cases.

We explore two extensions to the baseline model. The first expands brokers' bargaining space from decisions on forming coalitions to a transferable utility game, permitting unbounded transfers to incentivize or disincentivize coalition formation. Coalition formation decisions then reflect the maximization of aggregate broker profit. Importantly, insufficient tokenization can still occur due to tokenization's positive externality to investors on the largest island. Thus, a role for policy—specifically, fiscal transfers between investors and brokers—remains.

Our second extension pinpoints the impact of broker heterogeneity by developing a comparator case with one investor on each island. The equilibrium continues to feature excessive investment and insufficient tokenization, but an interoperability mandate alone now suffices to guarantee the social optimum. The need for subsidization only arises with broker heterogeneity because it is the *difference* in market power that permits the largest broker to earn excess markups and creates a wedge between aggregate broker profits and social welfare.

Related literature. We contribute to the literature by providing the first formal analysis of the *formation* of tokenized markets. This is distinct from the existing theoretical work that assesses the optimal design of such markets. In Chiu and Koeppl (2019), transaction validation gains credibility when there are more validators, which however depends on rewards that increase as the congestion of the tokenized market rises and its settlement speed declines—a consideration specific to public blockchains, not the private ledgers that are our focus.[6] Lee, Martin and Townsend (2024a,b) highlight a trade-off between the benefits of simultaneous settlement and the costs of a liquidity hold-up problem from instantaneous settlement. The latter originates in the requirement for the seller to own the asset at the time of trade, which reveals information that can be exploited by the buyer. Both papers analyze the optimal design of a tokenized market in view of this trade-off: Lee, Martin and Townsend (2024a) focus on the joint design of settlement and trading systems, while Lee, Martin and Townsend (2024b) focus on program execution and transaction verifiability.[7]

Our paper also relates to the literature on the formation of financial market infrastructures. Dugast, Üslü and Weill (2022) are closest to us within this literature, modeling bro-

[6] Validation effort depends on the extent of trust: in a trusted (permissioned) environment a small number of institutions validate each other's transactions using consensus mechanisms that are much faster and cheaper than the alternatives commonly used in trustless environments on public blockchains. On consensus mechanism trade-offs, see also Budish (2025) and Eidan et al. (2026).

[7] Besides financial asset trade, programmable tokens have also been studied in relation to non-financial assets (Chemla and Tinn, 2026; Harvey et al., 2026; Laschinger et al., 2024), composed financial products (Lee, 2026; Rostek and Yoon, 2024), financial stability (Camera and Gioffrè, 2026; Georgiadis-Harris, Guennewig and Mitkov, 2024; Lin et al., 2026), payments (Adrian et al., 2023, 2022; Chiu and Monnet, 2026; Kahn and van Oordt, 2022; Lee and Martin, 2026), contract negotiation and enforcement (Brunnermeier and Payne, 2023; Brzustowski, Georgiadis-Harris and Szentes, 2023; Cong and He, 2019; Duffie and Wang, 2026; Townsend and Zhang, 2023), platform governance (Abadi and Brunnermeier, 2024; Auer, Monnet and Shin, 2025; Camera, Charness and Chemaya, 2026; Cong et al., 2025b; Garratt and Monnet, 2023; Li and Mann, 2025; Reuter, 2024; Sockin and Xiong, 2023), and monetary policy (Project Pine, 2025).

kers of heterogeneous trading capacity that choose between centralized and OTC markets.[8] Small brokers prefer the OTC market due to the risk sharing that large brokers can profitably provide to them there, while mid-sized intermediaries opt for the centralized market. This resembles the emergence of a partial coalition between the largest and smallest brokers in our equilibrium. However, in Dugast, Üslü and Weill (2022) neither market enables exclusion, as brokers are free to join either. Brokers thus have two pre-existing, open-access markets with different features to choose from—put differently, the framework is closer to our model with: 1) costless creation of the tokenized market; 2) frictionless conversion between tokenized and legacy trade; 3) an interoperability mandate already in place.[9]

The remainder of the paper is organized as follows. The next section presents the setup of the model. Section 3 derives the private sector equilibrium, then Section 4 assesses welfare and Section 5 analyzes socially optimal policy. Section 6 discusses the model extensions and Section 7 concludes. Proofs can be found in the Appendix.

2 Model

Our model contains two types of agents: investors and brokers. Investors cannot trade with each other directly: to trade, they must engage the services of a broker. Investors and brokers are distributed across three islands, each containing one local broker and a population of investors: one investor on Island 1, two on Island 2, and three on Island 3. We denote investors by their location (island) l and a within-island identifier i, such that $li \in \{11, 21, 22, 31, 32, 33\}$. Tables 1 and 2 summarize the timing of events and the costs facing brokers; the next sections explain these in detail. All agents are risk neutral and fully informed about the structure of the game.

[8] Other papers in this literature analyze the emergence of market structure under different forms of heterogeneity, including heterogeneity in investors' asset valuations (Babus and Parlatore, 2022), wealth (Koeppl, 2012), risk preferences (Malamud and Rostek, 2017), search abilities (Lu, Puzzello and Zhu, 2025), and trading intensity (Farboodi, Jarosch and Shimer, 2023; Sambalaibat, 2025; Üslü, 2019), as well as endogenous market entry of investors (Chen and Duffie, 2021; Duffie, Dworczak and Zhu, 2017; Lee and Wang, 2025; Pagano, 1989) and intermediaries (Atkeson, Eisfeldt and Weill, 2015; Biais, 1993; Bolton, Santos and Scheinkman, 2016; Chang and Zhang, 2021; Chiu, Eisenschmidt and Monnet, 2020; Farboodi, 2023; Farboodi et al., 2019; Miao, 2006; Rust and Hall, 2003; Wang, 2016).

[9] Exclusive coalition formation also distinguishes us from papers on externalities in financial technology adoption (Alvarez et al., 2026; Crouzet et al., 2026; Crouzet, Gupta and Mezzanotti, 2023; Higgins, 2024; Pagnotta and Philippon, 2018), including papers on interoperability among payment providers (Alok et al., 2024; Bianchi et al., 2023; Bianchi and Rhodes, 2024; Bouvard and Casamatta, 2024; Brunnermeier, Limodio and Spadavecchia, 2023; Casiano, 2026; Chiu and Wong, 2022; Copestake et al., 2025; Frost et al., 2025).

Table 1: Timing of the game

	Investors	**Brokers**
Stage 0	Born with asset & money	—
Stage 1	—	Negotiate coalition: members become **token** brokers, rest remain **legacy**
Stage 2	—	Post fees
Stage 3	Choose broker (at cost τ if off-island); deposit asset	Receive assets: **tokenize** / **hold**
Stage 4	Preference shocks realize	—
Stage 5	Trade if matched	Sequence & settle trades

Table 2: Broker costs

	Legacy broker	**Token broker**
Market setup (Stage 1, one-off)	0	s/m
Settlement (Stage 5, per trade)		
Intra-broker	0	0
With **legacy** broker	γ	$\gamma + \kappa$
With **token** broker	$\gamma + \kappa$	$\varepsilon \to 0$

2.1 Investors

Aside from their locations across islands, investors are initially identical. All investors are born with one unit of a homogeneous, indivisible asset, plus a money endowment η that they can use for payments. At an intermediate stage of the game, investors receive a preference shock: with 50% chance an investor is assigned type $t = a$ and wants to sell her asset; with 50% chance she is assigned type $t = b$ and wants to buy one additional asset.[10] We normalize the gains from trade to 1: type a nets a (consumption-equivalent) utility gain of 1 from selling the asset and type b gains the same amount from buying the asset.[11]

Investors require brokers to trade. Before the shock, each investor therefore chooses a broker at which to deposit her asset and so becomes a client of the broker. After the shock, the broker can—on the investor's instruction—put the asset up for sale or request the purchase of an additional asset. Depositing an asset with the broker on the investor's native island (the "local" broker) is costless, but becoming a client of a broker on another island incurs a non-pecuniary distance cost τ. This endows each broker with a degree of market power over the investors on its island.[12]

Brokers charge fees to facilitate trades on behalf of investors. Numbering brokers by the island on which they are located, broker $j \in \{1, 2, 3\}$ charges a fee f_{lj} to facilitate a match for an investor from island l. Any part of the investor's endowment η that is not used for fees or trades enters their utility linearly (i.e., preferences are quasilinear in η).

The expected utility u^e_{li} of an investor li choosing broker j is thus given by

$$u^e_{li} = \eta + \max\{(1 - f_{lj}) \left(\tfrac{1}{2}P_{aj} + \tfrac{1}{2}P_{bj}\right), 0\} - \tau_{lj} \tag{1}$$

where P_{tj} is the probability that an investor of type t is matched with her opposite type, conditional on choosing to trade through broker j, and the max operator reflects that investors

[10] These investor types, $t \in \{a, b\}$, are a stylized representation of any source of ex-post heterogeneity that generates differences in the valuations of an asset, and hence a desire to trade. For instance, type a could reflect investors hit by a liquidity shock that makes them willing to sell their asset to type b investors at a discount to its held-to-maturity value—as is standard in the literature on financial market structure (Duffie, Gârleanu and Pedersen, 2005; Hugonnier, Lester and Weill, 2019; Maurin, 2022).

[11] We do not explicitly model investor bargaining on the shares of gains from trade (i.e., through asset pricing), since it is not essential to our analysis. We also note that the equal probability of drawing each type is without loss of generality, since any ex-post set of investor types can materialize (Table A.1).

[12] In practice, such market power can stem from broker-client relationship value and τ then captures a switching cost, where an island's population is taken to represent an initial clientele. Broker-client relationship value is well documented (Allen and Wittwer, 2023; Di Maggio, Egan and Franzoni, 2022). Industry estimates also indicate that investors prefer having one financial services provider due to relationship value.

will not place a trade unless the net return from doing so is weakly positive. Lastly,

$$\tau_{lj} = \begin{cases} 0 & \text{if} \quad l = j \\ \tau & \text{if} \quad l \neq j, \end{cases} \tag{2}$$

that is, the investor incurs a cost of τ when choosing an off-island broker.[13]

2.2 Brokers

Brokers earn fee revenue from facilitating investors' trades, as described above, but may also incur settlement costs. These costs depend on brokers' technology. We first describe the available settlement technologies and their costs, then we turn to broker fees and profits and describe how brokers determine whether to form tokenized markets.

2.2.1 Settlement technologies and costs

Table 2 shows the relationship between brokers' trading technology and the costs of settlement. There are four possible cases.

First, if a given broker's clientele includes some investors of type a and some investors of type b, then the broker can match pairs of opposing types and execute trades.[14] Such intra-broker matching is costless, as shown in the first settlement row of Table 2. Costs only arise when there is a need to resolve settlement risk among brokers.

The second case is where both brokers participating in an inter-broker trade use only the legacy technology. We assign a cost $\gamma > 0$ to overcoming inter-broker settlement risk in this case—for example, by using a clearinghouse.

The third case is where both brokers use the same tokenized market. In this case, settlement risk is resolved by programs that lock in an asset and a payment for exchange as soon as a trade is agreed upon. Settlement on such markets is therefore almost costless—almost, because validating transactions on a private ledger involves a positive but small validation cost (as noted in footnote 6) unlike intra-broker settlement where no validation is needed. We represent this cost by $\varepsilon \to 0$, taking the cost as zero throughout our analysis but utilizing ε as a tie-breaker—i.e., all else equal, a broker will prefer intra-broker over

[13] We do not incorporate remitted broker profits in investor utility. Note that in Section 4, the policymaker does weigh both investor utility and broker profits.

[14] In cases where there are multiple possible investor pairings on a given market, we assume that a random draw determines the allocation of trades. For instance, if on a given market (e.g., an intra-broker market) brokers represent one type a investor and three type b investors, then each type b investor has a one-third chance of being matched with the type a investor.

tokenized-market settlement.[15]

The final case is where one broker has tokenized the assets it has received and the other broker uses only the legacy technology. While tokenization leads to efficiency improvements when settling trades among token brokers, it can also lead to efficiency losses for trade with remaining legacy brokers. To sell a tokenized asset on the legacy market, a broker must destroy the token and request that the custodian of the underlying asset releases the asset for trade. Subsequently, the released asset is exchanged for money on the legacy market at transaction cost γ. Detokenizing the asset takes time, raising the (opportunity) cost of settlement by $\kappa > 0$.[16]

This simple cost structure highlights that tokenization is fundamentally a *joint* technology adoption decision among brokers. A given broker can only reap efficiency gains from tokenizing its investors' assets if at least one other broker also adopts the same technology. A single broker tokenizing its investors' assets cannot achieve any cost savings from doing so, since intra-broker trade is always costless.

We refer to two or three brokers jointly adopting tokenization technology as the creation of a tokenized market between them. Creating a tokenized market comes with total setup cost s to the involved brokers, and we assume that this cost is evenly split among them. Moreover, such tokenized markets are excludable: when two brokers form a tokenized market on a private ledger, the excluded broker cannot join them by tokenizing on its own.[17]

2.2.2 Broker fees

Brokers earn revenues from transaction fees $f_{lj} > 0$ on successful matches. We allow brokers full freedom to price discriminate by setting different fees based on an investor's origin l. For example, Broker 1 can set one fee for investors on its own island, one fee for investors from Island 2, and one for investors from Island 3. Each broker j thus posts a menu of fees $\boldsymbol{f}_j = (f_{1j};\, f_{2j};\, f_{3j})$ that investors observe before choosing a broker.[18]

Brokers engage in Bertrand competition on fees subject to up to three sources of differ-

[15] For more detail on brokers' preference for internal settlement, see Bowman, Huh and Infante (2024) and Lu, Macchiavelli and Wallen (2023).

[16] This argument applies equivalently for buying an asset on a legacy market and then tokenizing it.

[17] We model a binary broker technology choice: brokers either (jointly) become token brokers, which tokenize any assets they receive from investors, or remain legacy brokers. Our setup thus abstracts from brokers that tokenize a subset of their assets. Tracking *how many* assets each broker tokenizes would compromise tractability for little qualitative benefit: investor type realizations at Stage 4 would lead to ex-post suboptimal numbers of tokenized assets, implying higher expected settlement costs under partial coalitions compared to the grand coalition, thus reproducing our central trade-off between trade efficiency and trade diversion from tokenization.

[18] Interpreting our island model as attached clientele, price discrimination is in line with the sweeteners that brokers offer new clients in practice.

entiation among them. The first stems from positive τ, which provides a broker with market power when setting fees for its local investors. Broker 3 has the most market power—since it has the most on-island investors—and Broker 1 has the least. The second source of differentiation is variation among brokers in the probabilities P_{kj} that their clients obtain a match, and hence gains from trade. Such variation arises when intra-broker markets differ in size (that is, brokers attract different numbers of clients) or when a tokenized market forms among a subset of brokers. The third source of differentiation is heterogeneity in the brokers' expected marginal cost of facilitating trades, which derives from differences in brokers' expected distribution of trades across markets that have different costs of trade settlement.

2.2.3 Broker profits

Let $n_{lj,p}$ denote the number of trades facilitated by broker j for investors from origin l on market $p \in \{I, T, L\}$, representing the intra-broker, tokenized and legacy markets respectively. Furthermore, $n_{lj} = \sum_p n_{lj,p}$ denotes the total number of trades, across all three markets, facilitated by broker j for investors from origin l. We stack these totals from all three origins in the vector $\mathbf{n}_j = (n_{1j}; n_{2j}; n_{3j})$. Lastly, $n_{j,L} = \sum_l n_{lj,L}$ denotes the total number of trades facilitated by broker j on the legacy market, across investors from all three origins. The broker's total profit is then:

$$\pi_j = \boldsymbol{f}_j' \mathbf{n}_j - (\gamma + \mathcal{T} \cdot \kappa) \cdot n_{j,L} - s_j \ . \tag{3}$$

Here, $\mathcal{T}$ is a dummy that takes value 1 if any tokenized market has formed at Stage 1 and 0 otherwise. This reflects that: (i) if no tokenized market has formed, all brokers use the legacy technology and κ is never paid; (ii) if a partial coalition formed at Stage 1, then any trade with the excluded legacy broker costs $\gamma + \kappa$; and (iii) if a coalition among all brokers formed at Stage 1, then $n_{j,L} = 0 \ \forall j$ and therefore $(\gamma + \mathcal{T} \cdot \kappa) \cdot n_{j,L} = 0$. The final term is

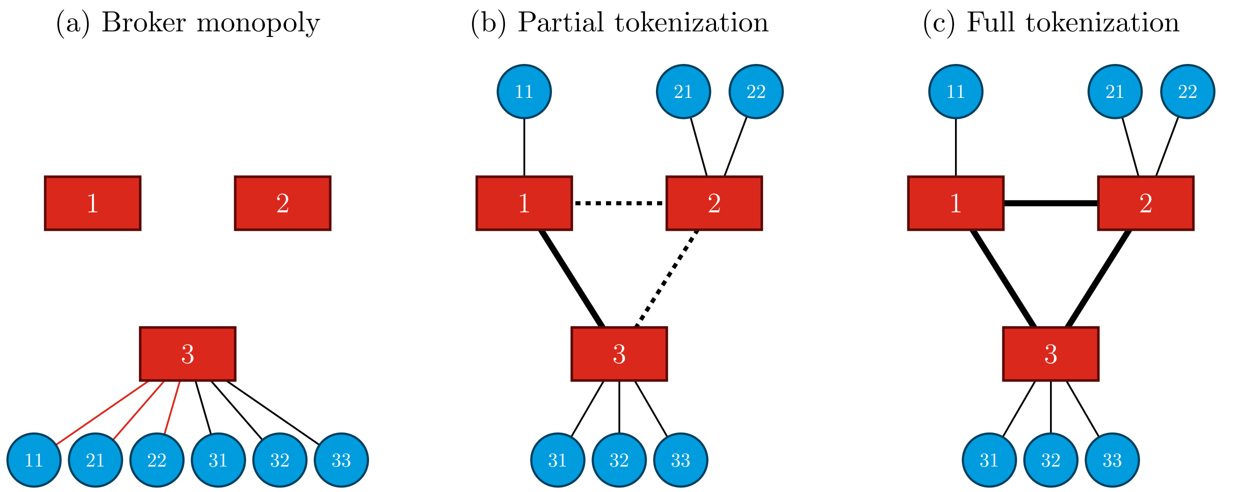
$$s_j = \mu_j \frac{s}{m} \tag{4}$$

and reflects the potential cost to broker j of forming a tokenized market, where m is the number of brokers participating and μ_j is a dummy taking value one if broker j is involved in forming such a market and zero otherwise.

2.2.4 Broker coalition negotiations

At Stage 1, coalitions of brokers have an opportunity to form a tokenized market. We allow for an arbitrarily long period of negotiation, and simply require that, for a coalition structure

Figure 2: Example outcomes



Notes: This figure shows some example permutations of (i) investors' choices of broker, and (ii) brokers' choices regarding tokenized market formation. Blue circles represent investors and red rectangles represent brokers. Intra-broker trading relationships are shown by thin solid lines, with red lines denoting non-local investor-broker pairs, for which the investor incurs a distance cost. Dotted lines show inter-broker relationships in the pre-existing legacy market, and thick solid lines show inter-broker relationships in a tokenized asset market.

that emerges to be an equilibrium, no broker or coalition of brokers would strictly prefer to deviate from their existing coalition.

Formally, denote by $J = \{1, 2, 3\}$ the set of brokers in the market and denote by $\mathcal{P}$ the set of all possible coalition structures, i.e., all partitions of J.[19] Each coalition structure $\mathcal{C} \in \mathcal{P}$ is associated with a single vector $\boldsymbol{\pi}^{\boldsymbol{e}}(\mathcal{C}, s, \gamma, \kappa) \in \mathbb{R}^3$ of expected payoffs $\pi_j^e(\mathcal{C}, s, \gamma, \kappa)$ for each broker j, given observed cost parameters s, γ and κ, where expectations are formed over the distribution of investor types at a subsequent stage. A coalition structure $\mathcal{C}$ is then an equilibrium if there does not exist any alternative coalition structure $\mathcal{C}'$ containing a new coalition $C' \notin \mathcal{C}$ such that $\pi_j^e(\mathcal{C}', s, \gamma, \kappa) > \pi_j^e(\mathcal{C}, s, \gamma, \kappa)\ \forall\ j \in C'$.[20] A tokenized market is formed when an equilibrium contains a coalition of more than one broker. For convenience, we attach the terms "No Coalition" and "Grand Coalition" to, respectively, the coalition structures $\{\{1\}, \{2\}, \{3\}\}$ and $\{1, 2, 3\}$.

Ultimately, which markets are active in the trading stage (Stage 5) depends both on

[19] Here, $\mathcal{P} = \{\{\{1\}, \{2\}, \{3\}\}, \{\{1, 2\}, \{3\}\}, \{\{1, 3\}, \{2\}\}, \{\{1\}, \{2, 3\}\}, \{\{1, 2, 3\}\}\}$: the first of these is the case where no tokenized market forms, the second to fourth structures represent the partial coalitions of two brokers, and the last structure is the three-broker coalition.

[20] For instance, the structure containing the three-broker coalition ($\mathcal{C} = \{\{1, 2, 3\}\}$) is an equilibrium if: (i) no individual broker would strictly prefer to use only the legacy market for inter-broker trade, and (ii) no pair of brokers would both strictly prefer to form their own tokenized market that excludes the third broker.

the coalitions that form at Stage 1 and the sorting of investors at Stage 3. To fix ideas, Figure 2 presents three (among the many) possible configurations. Figure 2a depicts a case in which all investors choose the same broker. All trading relationships are then intra-broker (shown by thin solid lines), with some investors having incurred extra costs to deposit their asset off-island (shown by red lines). In Figures 2b and 2c, all investors choose their local broker. Intra-broker trade can still occur at either Broker 2 or Broker 3. Inter-broker trade may occur too, either through the legacy market (shown by dotted lines) or through the tokenized market (shown by thick solid lines). Figure 2b shows the case where both types of inter-broker trade coexist, as only Brokers 1 and 3 form a tokenized market. In Figure 2c, instead, all brokers join the tokenized market, so no trade occurs using the legacy technology.

2.3 Optimization problems

To prepare for solving the game by backward induction, we present the agent optimization problems in the reverse order of the decision stages in Table 1. Stage 4 is the type realization stage, and therefore Stages 1-3 and 5 are, respectively, ex-ante and ex-post decision stages.

At Stage 5, coalition structures, broker fees, and the sorting of investor types across brokers are all known. Brokers' decisions at that stage are therefore simple: they minimize settlement costs. All brokers prefer to settle any feasible trades intra-broker, since such trades are cheapest (Table 2). Among the remaining feasible trades, token brokers prefer trading with each other over trading with legacy brokers. Thus, coalition structures produce the sequences of trade settlement shown in Table 3.

Under No Coalition, all inter-broker trade occurs on the legacy market and thus any feasible trades that brokers cannot settle intra-broker are settled on that market. Under the Grand Coalition, the same applies except that the tokenized market is then the only inter-broker market. Under any partial coalition, there are two token brokers, who prefer settling inter-broker trades with each other to trading with the one remaining legacy broker. Hence, after settling feasible trades intra-broker, the token brokers subsequently settle remaining feasible trades with each other and only turn to the expensive legacy market as a last resort.

Table 3: Trade settlement sequencing emerging at Stage 5.

No Coalition	**Grand Coalition**	**Partial coalition**
1. Intra-broker	1. Intra-broker	1. Intra-broker
2. Legacy market	2. Tokenized market	2. Tokenized market
		3. Legacy market

At Stage 3, each investor li chooses her broker j_{li} to maximize the expected value of her payoff u^e_{li}, taking into account the (observed) coalition structure $\mathcal{C}$ and the (observed) fees of all brokers $\boldsymbol{f} = (\boldsymbol{f}_1;\ \boldsymbol{f}_2;\ \boldsymbol{f}_3)$:

$$\max_{\jmath_{li}} u^e_{li}(\mathcal{C}, \boldsymbol{f})\,. \tag{5}$$

At Stage 2, each broker j sets its menu of fees $\boldsymbol{f}_j$ to maximize its expected profits:

$$\max_{\boldsymbol{f}_j} \pi^e_j(\mathcal{C}, \boldsymbol{f}) \tag{6}$$

where the broker takes into account the observed coalition structure $\mathcal{C}$ from Stage 1 and foresees how its fees will subsequently affect investors' choices of broker at Stage 3.[21]

Lastly, at Stage 1 brokers backward induce all subsequent stages of the game and bargain over coalition structures $\mathcal{C}$ in an attempt to maximize their expected profits.

2.4 Constraints

We aim to analyze the incentives of competing brokers, each with a degree of market power, to jointly form a tokenized market. With this in mind, we center attention on cases that do not produce broker monopolies. Such monopolies arise at two opposite extremes of our setup: when distance costs are so large that investors are walled in at their local broker, and when such costs are so small that investors all converge on a single broker.

2.4.1 Three uncontested monopolies

If distance costs are so large that brokers become uncontested monopolists with respect to their local investors, brokers set fees equal to investors' gains from trade, regardless of the coalition structure. The constraint $\tau \leq \frac{2}{3}$ suffices to exclude uncontested monopolies.

2.4.2 Single monopolist

We aim to spotlight cases where the market for brokerage is somewhat contestable, but not so contestable that brokers' incentives shift away from coalition formation toward intra-broker unification. This is essentially another form of monopolization, this time by attracting all investors to the same broker (as in Figure 2a, for example). We therefore make two further assumptions.

[21] Writing equation (3) in expected terms yields $\pi^e_j(\mathcal{C}, \boldsymbol{f}) = \boldsymbol{f}'_j \mathbf{n}^e_j(\mathcal{C}, \boldsymbol{f}) - (\gamma + \mathcal{T} \cdot \kappa) \cdot n^e_{j,L}(\mathcal{C}, \boldsymbol{f}) - s_j$, where the expectations operator on $\mathbf{n}^e_j$ and $n^e_{j,L}$ reflects that investor types—drawn at Stage 4—affect matching at Stage 5.

First, we impose that $\tau \geq \frac{1}{3}$. This lower bound on τ ensures that the equilibrium wherein all investors choose their local brokers is always feasible, where feasibility here means that investors' priors are self-confirming (i.e., given that each investor believes that every other investor will choose their local broker, each investor optimally chooses their local broker).

Second, we assume that whenever the equilibrium wherein all investors choose their local brokers is feasible, it comes about. Investors' selection of brokers at Stage 3 can produce multiple equilibria because investors impose externalities on each other: the more investors pool on one broker, the more attractive that broker is to other investors, given that brokers prioritize intra-broker trades.[22] Our assumption equates to using a minimal-movement prior as an equilibrium selection criterion in such cases. The notion that agents expect "stickiness" of other agents—i.e., status-quo bias—is well established (Battigalli et al., 2015; Fudenberg and Levine, 2016; Guney and Richter, 2018).[23]

2.4.3 Other constraints

To facilitate the derivation of analytical solutions for optimal broker fees, we constrain the maximum cost of clearing legacy market transactions to be no more than one-sixth of an investor's gains from trade: $0 < \gamma + \kappa \leq \frac{1}{6}$.[24] Moreover, we restrict the size of the token-to-legacy settlement friction to be smaller than the overall friction of settlement on legacy markets: $0 < \kappa < \gamma$. Under this assumption, κ is never so large that it prevents feasible token-to-legacy trades from occurring. Furthermore, to ensure that investors have enough endowment to cover fees and retain positive utility if they incur non-pecuniary cost τ, we set $\eta \geq 2$. Finally, to resolve multiplicity on indifference thresholds, and without loss of generality, we introduce tie-breaking assumptions in favor of (larger) tokenized market formation when brokers are exactly indifferent between coalitions. Overall, our parameter constraints can then be summarized as:

$$\tau \in \left(\frac{1}{3}, \frac{2}{3}\right); \quad \kappa \in (0, \gamma); \quad (\gamma + \kappa) \in \left(0, \frac{1}{6}\right); \quad \eta \geq 2 \tag{7}$$

[22] For example, for a given parameterization the following equilibria could co-exist: first, an equilibrium where each investor acts from the prior that all other investors will choose their local brokers and this prior is confirmed by each investors' decision, as in Figures 2b and 2c; second, an equilibrium where each investor expects all other investors to choose Broker 3 and, given this prior, all investors indeed aggregate at Broker 3, as in Figure 2a.

[23] A minimal-movement prior also relates to the rise of passive investing (Jiang, Vayanos and Zheng, 2025; Malhotra, 2024)—a given investor acts from the prior that other investors are passive.

[24] This incurs no qualitative loss of generality, as seen in Section 4 where all relevant policy zones are represented.

3 Equilibrium

This section presents the equilibrium of the game described in Section 2, which Lemma 1 shows depends on the tokenized market setup cost s, the legacy market trading cost γ, and the token-to-legacy settlement friction κ.[25]

Lemma 1 (Baseline equilibrium) *If $s > \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$, no tokenized market forms in equilibrium. If $s \leq \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$, Brokers 1 and 3 form a tokenized market.*

Proof of Lemma 1. See Appendix A.1. ■

Figure 3a depicts these regions in the (s, γ) space, while depicting κ as a shifter: the blue line is drawn for $\kappa \to 0$ and we first discuss this case, before turning to the impact of κ. Above the blue line, no multi-broker coalition forms—i.e., no tokenized market is created—because the setup cost is too high relative to the potential benefits. On and below the blue line, Brokers 1 and 3 form a tokenized market, excluding Broker 2. This outcome reflects two forces. First, tokenized markets in general are more attractive because, for any given efficiency gain γ relative to the legacy market, their setup cost s is lower. Second, forming a partial coalition with two brokers diverts trade from the excluded broker toward the participating brokers. To see this, note that the 1 & 3 Coalition forms even as γ approaches zero, as long as s is not too high. Here, forming a tokenized market offers no reduction in costs—since all trades are already costless—but it still provides a benefit to participating brokers in the form of preferential access to each other's investors, due to the earlier execution chosen for trades on the tokenized market (Table 3). This raises the matching probabilities offered by the participating brokers, at the expense of the excluded broker, in turn allowing them to charge higher fees.

Among the three feasible partial coalitions (1 & 2, 1 & 3, 2 & 3), why does the 1 & 3 Coalition form? Brokers 2 and 3 are each other's fiercest competitors: their larger local clienteles mean that, ceteris paribus, they offer better matching odds than Broker 1, due to their greater opportunities for intra-broker clearing. Thus, in the competition for Broker 2's local clients, Broker 3 binds how high Broker 2 can set its fees given distance costs and, similarly, Broker 2 binds Broker 3's fees. Forming a partial coalition with Broker 1 gives each the opportunity to weaken the other's position, since an excluded broker offers a worse matching probability to investors. Hence, when either Broker 2 or 3 joins a partial coalition with Broker 1, it expects to process more trades and at higher equilibrium fees. Broker 1 also benefits from this trade diversion, and benefits most when forming a partial coalition

[25] Parameters η and τ do not influence equilibrium outcomes given the constraints in (7).

with Broker 3, because this gives it the best chance to match its investor on the tokenized market.

Lemma 1 also reveals the impact of the token-to-legacy settlement friction: for given values of s and γ, a larger κ reduces the attractiveness of forming a partial coalition (the blue line shifts down). Intuitively, a higher cost of trading with the excluded broker reduces the benefits to Brokers 1 and 3 of forming a tokenized market, so they will only do so when compensated by a lower setup cost s and/or a larger relative efficiency gain γ.

4 Welfare

Denote the total expected profit of the brokerage sector by $\Pi^e = \sum_{j=1}^{3} \pi_j^e$, and the total expected utility of the investors by $U^e = \sum_{li} u_{li}^e$. Total expected welfare is then given by $W^e = \Pi^e + U^e$. For convenience, we let "welfare" refer to total *expected* welfare.

Lemma 2 (Optimal outcomes) *If $s > \frac{13}{8}\gamma$, welfare is maximized when no tokenized market exists. If $s < \frac{13}{8}\gamma$, welfare is maximized by a tokenized market that includes all brokers. If $s = \frac{13}{8}\gamma$, welfare is equal under both outcomes.*

Proof of Lemma 2. See Appendix A.2. ■

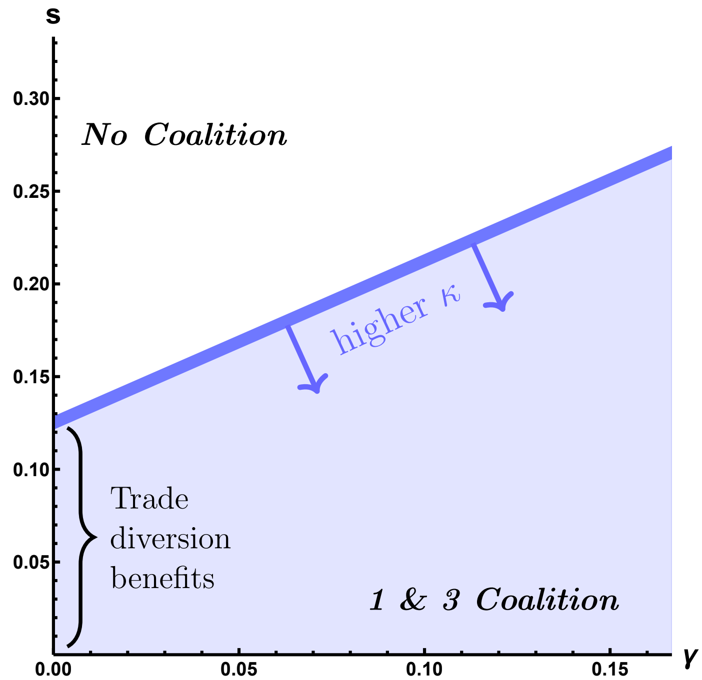
The red dashed line in Figure 3b plots the boundary in Lemma 2. Below the line, the Grand Coalition is socially optimal because (i) the efficiency benefits of tokenization outweigh the costs and (ii) the gap between these benefits and costs is largest in the Grand Coalition, where costly legacy market trade is fully eliminated. Above the line, setup costs are too high and No Coalition is socially optimal. The line passes through the origin because without efficiency gains any setup cost is too high, and without a setup cost any efficiency gains justify creating the Grand Coalition.[26] Importantly, investor utility makes the gradient of the red dashed line steeper than if only broker profits entered welfare, because part of the efficiency gains from tokenization endogenously pass through to investors in the form of lower equilibrium fees.[27]

[26] We note that κ, η and τ do not enter the expression in Lemma 2. For κ, this is because partial coalitions—where κ is incurred—are always welfare dominated and the comparison for social optimality thus centers on No and Grand coalitions. Moreover, welfare trivially increases for a higher η, but this does not affect the welfare comparison between coalitions. Lastly, τ is not incurred in equilibrium since all brokers serve their local investors (but τ does affect the distribution of welfare between investors and brokers because a larger τ allows brokers to extract higher fees from their local investors).

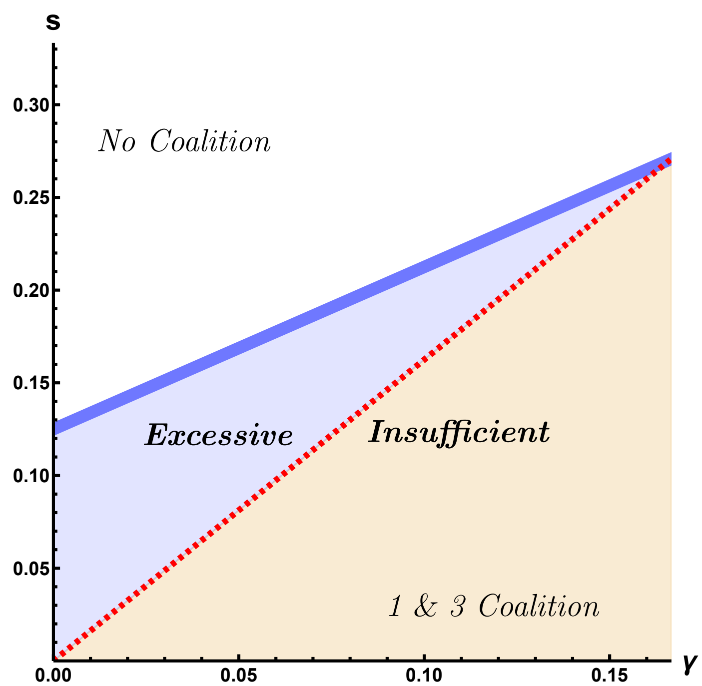
[27] These investor benefits from tokenization are discussed in detail Section 5.

Figure 3: Equilibrium without policy

(a) Baseline



(b) Excessive investment and insufficient tokenization



Notes: These figures depict the baseline equilibrium as a function of the tokenized market setup cost s and the legacy market trading cost γ. In Panel (a), above (below) the thick blue line the equilibrium features No Coalition (the 1 & 3 Coalition). The blue line is drawn for $\kappa \to 0$ and an increase in κ shifts this line down. In Panel (b), below the thick blue line and to the left (right) of the dotted red line, the equilibrium reflects excessive (insufficient) coalition formation.

Proposition 1 (Excessive investment & insufficient tokenization) *The private equilibrium features excessive investment in tokenization if* $\frac{13}{8}\gamma \leq s \leq \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$, *and insufficient tokenization if* $s \leq \frac{13}{8}\gamma$.

Proof of Proposition 1. The result follows immediately from comparing the regions defined in Lemmas 1 and 2. ■

The shaded regions in Figure 3b illustrate the cases in Proposition 1. No Coalition is socially optimal everywhere above the red dashed line, but in the blue region brokers form the 1 & 3 Coalition. Trade diversion benefits create private incentives that result in excessive investment: s is incurred to set up a tokenized market, but welfare would be higher if instead brokers relied only on the legacy market. In contrast, in the yellow region it is socially optimal to incur s and set up a tokenized market, but the coalition that forms privately is too narrow: while Brokers 1 & 3 divert some trades from Broker 2, the remaining trades with that broker occur at the (socially wasteful) cost $\gamma + \kappa$.

Corollary 1 (Token-to-legacy friction and welfare) *A lower token-to-legacy settlement friction* κ *expands the region for which excessive investment occurs.*

Proof of Corollary 1. The result follows immediately from Proposition 1. ■

Lastly, we note that a lower friction on trade between tokenized and legacy markets exacerbate the excessive investment problem. The lower friction increases the payoff to Brokers 1 and 3 of forming a tokenized market, leading them to form a (socially costly) partial coalition in some cases where No Coalition would otherwise occur.

5 Policy analysis

A sufficiently powerful policymaker could straightforwardly attain the social optimum by forbidding tokenization above the red line in Figure 3b and mandating the Grand Coalition below it. However, political considerations often preclude such heavy-handed approaches. This section considers whether a policymaker can maximize social welfare by less intrusive means. In particular, we focus on the two main types of policy initiatives that have been pursued to date: an interoperability mandate and a public-private partnership to form a tokenized market. Among these two, the interoperability mandate is arguably the lighter-touch policy because it merely requires that brokers do not exclude others when creating a new ledger, whereas a public-private partnership involves using fiscal resources to help create the tokenized market. We therefore start our policy discussion from the interoperability mandate and then ask whether and when it should be supplemented by (the minimum necessary) public cost-sharing.

We incorporate the interoperability mandate—introduced prior to Stage 1—as a requirement that any coalition of two brokers deciding to form a tokenized market must offer the third broker the opportunity to join at no cost. Specifically, we break Stage 1 into the following three sub-stages. At Stage 1.1, each broker j sets a budget limit of s_j^{max} that it is willing to invest in tokenization. At Stage 1.2, brokers can jointly form a tokenized market by each investing $s_j \leq s_j^{max}$. If a partial coalition C forms at Stage 1.2, Stage 1.3 occurs, in which the excluded broker $k \notin C$ is offered the opportunity to join C at no cost (i.e., $s_k = 0$).[28] With this setup we derive the following results:

Lemma 3 (Interoperability mandate & excessive investment) *An interoperability mandate prevents the excessive investment that occurs in equilibrium when* $\frac{13}{8}\gamma \leq s \leq \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$.

Proof of Lemma 3. See Appendix A.3. ■

The interoperability mandate counteracts trade diversion: partial coalitions no longer arise in equilibrium because if the participating brokers paid $\frac{s}{2}$ to form a coalition at Stage 1.2, the excluded broker would join for free at Stage 1.3. Brokers thus choose between No Coalition and the Grand Coalition. In the blue region in Figure 4a at least one broker is never willing to form the Grand Coalition, so No Coalition results. The blue zone is where investment is excessive in the private equilibrium (as in Figure 3b), so in that region the interoperability mandate leads to the socially optimal outcome of No Coalition.

Lemma 4 (Interoperability mandate & insufficient tokenization #1) *An interoperability mandate prevents the insufficient tokenization that results in equilibrium if* $s \leq \frac{27}{20}\gamma$.

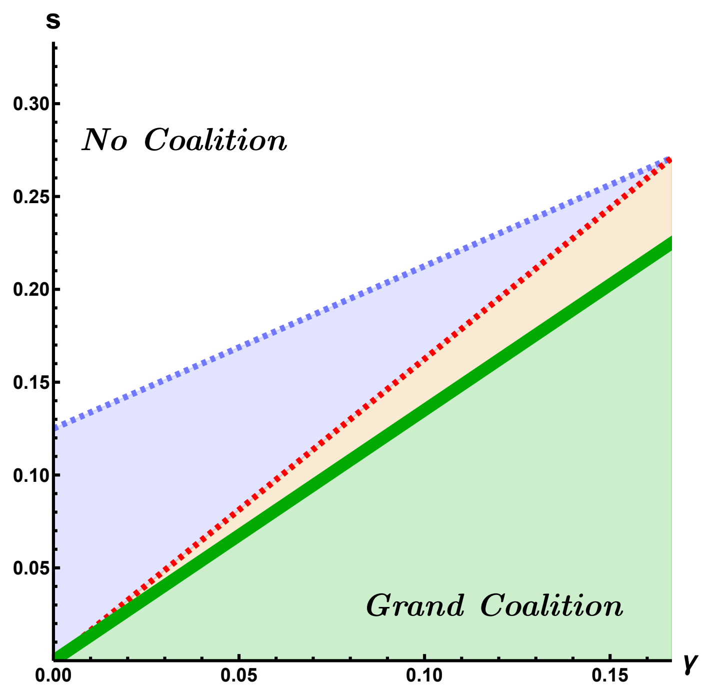
Proof of Lemma 4. See Appendix A.4. ■

This result relates to the green region in Figure 4a, where tokenization is insufficient in the private equilibrium (Figure 3b). The green line shows the boundary below which all brokers prefer the Grand Coalition to No Coalition.

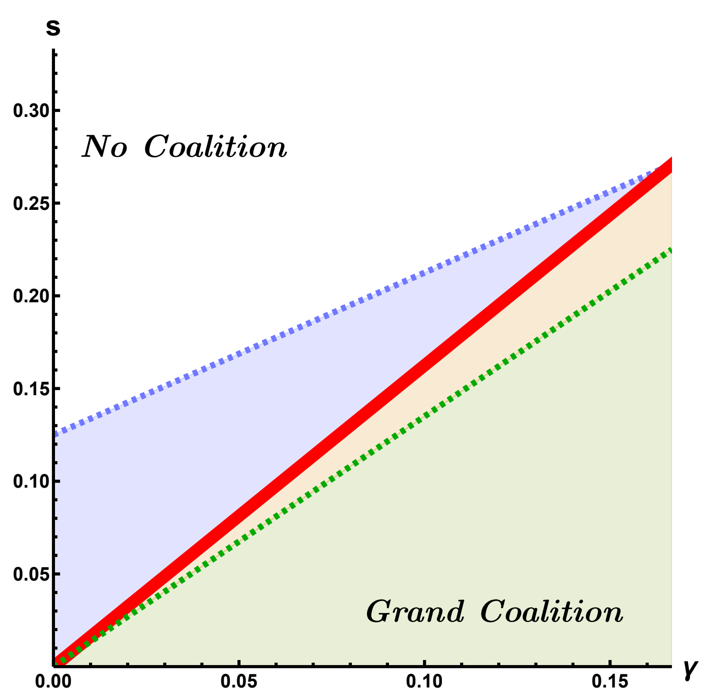
[28] We include Stage 1.1 to avoid a free-rider problem whereby each broker has an incentive to block the Grand Coalition at Stage 1.2 in the hope of forcing the other brokers to bear the setup cost for a tokenized market that it can then join for free in Stage 1.3. Including Stage 1.1 is not a departure from the baseline, since a model featuring Stages 1.1 and 1.2 alone (without Stage 1.3) would lead to the same outcomes as the baseline.

Figure 4: Equilibria with policy

(a) Equilibrium with an interoperability mandate



(b) Equilibrium with an interoperability mandate and tokenization subsidy



Notes: Panel (a) depicts the equilibrium of the model in the presence of an interoperability mandate. Above the thick green line this features No Coalition, while below the thick green line it features the Grand Coalition. Panel (b) depicts the equilibrium in the presence of both an interoperability mandate and (in the yellow region) a tokenization subsidy. Above the thick red line this equilibrium features No Coalition, while below the thick red line it features the Grand Coalition. These panels are drawn for the case $\kappa \to 0$ and remain qualitatively valid for larger κ.

Lemma 5 (Interoperability mandate & insufficient tokenization #2) *An interoperability mandate does not prevent the insufficient tokenization that results in equilibrium when* $\frac{27}{20}\gamma < s \leq \frac{13}{8}\gamma$.

Proof of Lemma 5. See Appendix A.5. ■

This result relates to the yellow region in Figure 4a and is of a different nature to that in the yellow region in Figure 3b, where a tokenized market does form but is suboptimal because one broker is excluded. Instead, in the yellow region in Figure 4a there is an underinvestment problem because no tokenized market forms. This underinvestment has two causes: tokenization's positive externalities to investors and the need for unanimity among heterogeneous brokers. We discuss these in turn.

First, brokers do not internalize that the efficiency gains from tokenization also benefit investors through lower fees in equilibrium. Specifically, this benefit applies to investors on Island 3. Under No Coalition, the fees of Brokers 1 and 2 on their local investors are a markup over τ, whereas Broker 3 includes an additional markup over γ in this fee.[29] This added term originates in the relative market power of Broker 3, whose investors are relatively sticky because of that broker's large intra-broker market and associated matching odds. When the marginal cost of trade settlement is eliminated under the Grand Coalition, Broker 3's investors no longer face this added markup factor—Brokers 1 and 2 then price-compete more forcefully due to the zero marginal cost on settling trades.

Second, private coalition formation decisions require unanimity among members, so depend on the profits of the broker that is least amenable to the Grand Coalition. That broker does not internalize the impact of its decision on the profits of the other brokers. Since brokers are heterogeneous, the least amenable broker's preference for Grand versus No Coalition does not align with aggregate broker profits or, indeed, overall social welfare.

To address the underinvestment problem, we next consider a combination of the interoperability mandate with public cost sharing, which we introduce in the form of a tokenization subsidy. This subsidy $\sigma > 0$ is announced before Stage 1 and reduces the setup cost from s to $s - \sigma$. The subsidy is funded with a lump-sum tax on all investor endowments. By itself (without interoperability) such a subsidy is not a useful policy. This can be seen from Figure 3b: a subsidy σ effectively replaces the outcome at each point (γ, s) with the outcome at the point $(\gamma, s - \sigma)$ vertically below. Below the blue line, this shift never changes the coalition structure—regardless of the size of σ—since the 1 & 3 Coalition always forms. Above the blue line, the equilibrium is already socially optimal and there is nothing for the subsidy to

[29] See equation (A.1.7) in the Appendix.

improve upon. Thus, no simple tokenization subsidy can achieve the social optimum when used alone.[30]

Combining an interoperability mandate with a tokenization subsidy instead allows the policymaker to sufficiently incentivize the marginal broker to join the Grand Coalition in the yellow region of Figure 4a. We derive the following result:

Lemma 6 (Interoperability mandate plus tokenization subsidy) *An interoperability mandate combined with a tokenization subsidy $\sigma \geq \frac{11}{40}\gamma$ prevents the insufficient tokenization that results in equilibrium when $\frac{27}{20}\gamma < s \leq \frac{13}{8}\gamma$.*

Proof of Lemma 6. See Appendix A.6. ■

In Figure 4a, all points in the yellow region are directly above a point in the green region where the interoperability mandate produces the Grand Coalition. Hence, in all such cases a sufficiently large subsidy produces the Grand Coalition by moving the equilibrium point from (γ, s) to $(\gamma, s - \sigma)$. In combination with Lemmas 3 and 4 above, we thus have:

Proposition 2 (Optimal policy) *A policymaker with the ability to impose an interoperability mandate and subsidize the creation of tokenized markets can always achieve the socially optimal outcome. Specifically, the policymaker can achieve this by:*

1. *Taking no action (laissez-faire) if $s > \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$,*
2. *Imposing an interoperability mandate if $s \leq \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$,*
3. *Subsidizing the formation of tokenized markets if $\frac{27}{20}\gamma < s \leq \frac{13}{8}\gamma$.*

Proof of Proposition 2. When $s > \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$, No Coalition maximizes welfare (Proposition 1) and laissez-faire is optimal. Policies 2 and 3 result from combining Lemmas 3, 4, and 6. ■

Figure 4b shows the equilibrium outcomes under this policy package. In the white region, No Coalition is both the private equilibrium and optimal, so the policymaker takes no action. In the blue region, the private equilibrium leads to excessive investment, but the interoperability mandate produces No Coalition, the optimal outcome. Below the red line, the Grand Coalition is optimal, and this is achieved by the interoperability mandate alone (in the green region) or in combination with a sufficiently large subsidy (in the yellow region).

[30] A sufficiently large tax ($\sigma < 0$, representing, e.g., stringent regulatory or licensing requirements) could prevent excessive investment (blue zone of Figure 3b) by offsetting the gains from trade diversion that otherwise motivate Brokers 1 and 3 to form a tokenized market. However, such a tax cannot resolve insufficient tokenization, as there is no point in Figure 3b at which brokers form the Grand Coalition.

Corollary 2 (Token-to-legacy friction and policy) *A reduction in the token-to-legacy settlement friction κ shrinks the laissez-faire zone and expands the parameter space within which an interoperability mandate is needed to attain the social optimum.*

Proof of Corollary 2. The result follows directly from Proposition 2. ■

Various regulatory initiatives aim to reduce token-to-legacy settlement frictions by facilitating a more rapid release of underlying assets from custodians (see, for example, BIS, 2024; IOSCO, 2025). Corollary 2 reveals that such declines in κ could actually worsen market fragmentation, if pursued in isolation. Our model suggests that, instead, optimal regulation should combine efforts to reduce κ with an interoperability mandate. The former policy makes tokenization by *some* brokers more likely; the latter policy ensures that *all* brokers, as well as investors, are able to benefit from the resulting efficiency gains.

6 Extensions

This section considers two variations of the baseline model. The first generalizes coalition formation by allowing brokers to make side-payments to one another. The second considers a setting with one investor on each island, which implies (ex-ante) broker homogeneity.

6.1 Allowing side-payments

In this extension, we allow brokers to make side-payments to one another during coalition formation.[31] For instance, Broker 2 can now pay Brokers 1 and 3 to form the Grand Coalition instead of excluding it by forming the 1 & 3 Coalition. Alternatively, Broker 2 could pay Brokers 1 and 3 not to form any coalition.

We define an outcome as a coalition structure and an associated trio of net side-payments between brokers. Such an outcome is an equilibrium if no broker or coalition of brokers would strictly prefer to deviate from their existing coalition to an alternative coalition in a new coalition structure that is supported by feasible side-payments.

We modify the tokenized market formation game by defining the vector $\mathbf{v} = (v_{12}; v_{13}; v_{23})$ as a series of net side-payments v_{jk} from each broker j to each other broker k.[32] An outcome $(\mathcal{C}, \mathbf{v})$ is then the pair of a coalition structure $\mathcal{C}$ with a trio of net side-payments $\mathbf{v}$. Each such pair is associated with a single vector of expected broker payoffs $\boldsymbol{\pi}^e(\mathcal{C}, \mathbf{v}, s, \gamma, \kappa) \in \mathbb{R}^3$,

[31] This also effectively relaxes the assumption of an equal distribution of setup costs among coalition members, since side-payments can deliver any distribution of the net gains from tokenization.

[32] We also denote the same net transfers measured in the opposite direction by $v_{kj} \equiv -v_{jk}$.

given observed cost parameters s, γ and κ. Such an outcome $(\mathcal{C}, \mathbf{v})$ is an equilibrium if there does not exist any alternative pair $(\mathcal{C}', \mathbf{v}') \neq (\mathcal{C}, \mathbf{v})$ where: (i) $\mathcal{C}'$ contains a new coalition $C' \notin \mathcal{C}$ such that $\pi^e_j(\mathcal{C}', \mathbf{v}', s, \gamma, \kappa) > \pi^e_j(\mathcal{C}, \mathbf{v}, s, \gamma, \kappa)\ \forall\ j \in C'$, and (ii) $\pi^e_l(\mathcal{C}', \mathbf{v}', s, \gamma, \kappa) > \pi^e_l(\mathcal{C}, \mathbf{v}, s, \gamma, \kappa)$ for all l for whom $\sum_k v'_{lk} > 0$.[33] Using this definition we derive the following result:

Proposition 3 (Welfare with side-payments) *With broker side-payments, insufficient tokenization can occur so the private equilibrium does not always maximize welfare.*

Proof of Proposition 3. See Appendix A.7. ■

Intuitively, the ability to make side-payments allows brokers to act collectively to maximize their joint surplus. Excessive investment never occurs, since any trade diversion incentives can be outweighed by side-payments from the broker that would otherwise lose out. However, brokers' joint profit maximization does not align with social welfare, due to the aforementioned positive externality to investors (on Island 3) from the formation of the Grand Coalition. Thus, even with fully flexible side-payments an underinvestment problem can persist and thus a policy response—whereby the policymaker essentially acts as a conduit for investor side-payments to brokers—remains necessary to guarantee the social optimum:

Proposition 4 (Policy implications of side-payments) *With broker side-payments, a sufficiently large tokenization subsidy can always maximize welfare.*

Proof of Proposition 4. See Appendix A.8. ■

6.2 Homogeneous brokers

This extension explores the role that broker heterogeneity plays in our results by analyzing an alternative setup in which brokers are ex-ante identical. In this extension, each island has a population of one investor. To preclude broker monopoly formation (as discussed in Section 2.4) in this setup, we tighten the constraints on τ from $\tau \in \left(\frac{1}{3}, \frac{2}{3}\right)$ in (7) to $\tau \in \left(\frac{3}{8}, \frac{5}{8}\right)$.

We solve for the private-sector equilibrium through backward induction in the same manner as in the baseline model. However, with ex-ante identical brokers, whenever one partial coalition is a feasible equilibrium, all partial coalitions are: we cannot pinpoint which

[33] Intuitively, equilibrium requires that no alternative outcome exists in which: (i) a broker or coalition of brokers C' chooses to deviate from their existing coalition, and (ii) any broker—including one excluded from C'—that makes net payments to the other brokers is willing to do so. Note also that this definition allows a range of equilibrium outcomes to exist for each coalition structure: a given $\mathcal{C}$ could be supported by many different vectors of side-payments $\mathbf{v}$, reflecting different ways of dividing the surplus created, conditional on no broker or coalition of brokers receiving so little or paying so much that they would prefer to deviate.

of these forms, unlike the 1 & 3 Coalition in the baseline equilibrium. Instead, we assume that one of the three feasible partial-coalition equilibria is randomly selected.

Proposition 5 (Homogeneous brokers.) *The equilibrium features insufficient tokenization if $s \leq \frac{3}{2}\gamma$ and excessive investment if $\frac{3}{2}\gamma \leq s \leq \frac{1-\kappa}{4} + \frac{3}{4}\gamma$, and therefore the excessive investment zone expands as κ declines. An interoperability mandate eliminates both excessive investment and insufficient tokenization and thus always achieves the social optimum.*

Proof of Proposition 5. See Appendix A.9. ■

The private sector equilibrium in this extension shares several key features with the outcomes of the baseline model: whenever a tokenized market forms at all, it is through a partial coalition that excludes one of the brokers; moreover, the three zones seen in Figure 3b—socially optimal absence of tokenization, excessive investment, and insufficient tokenization—continue to exist.

Optimal policy in this extension does differ from the baseline, however, because the interoperability mandate is sufficient to ensure the social optimum and a tokenization subsidy is never needed. In the baseline, an interoperability mandate produces an underinvestment problem because (i) brokers do not internalize the spillover benefits of tokenization for investors, and (ii) brokers differ in their payoffs from forming the Grand Coalition. With homogeneous brokers, cause (ii) is resolved by definition. On cause (i), recall that, in the baseline, this results from Broker 3 charging an additional markup over γ under No Coalition, a markup that is in turn eroded if the Grand Coalition forms. With homogeneous brokers, no broker has sufficient market power to charge such an additional markup under No Coalition, so the externality never arises.

7 Conclusion

As more brokers explore the tokenization of financial assets, policymakers seek to reap the potential benefits that this technology offers while mitigating the risk of market fragmentation. This paper examines the drivers of such fragmentation and provides the first formal framework for assessing optimal policy responses.

We model the endogenous joint adoption of tokenized trading technology by brokers already connected through a legacy trading platform. Brokers with heterogeneous market power compete to attract investors and execute their trades. Coalitions of brokers can invest in creating a tokenized market that enables cheaper settlement among coalition members, but tokenizing the underlying assets also adds friction to trading with excluded brokers.

Trade diversion incentives result in equilibrium coalition structures that can feature excessive investment or insufficient tokenization.

We then examine potential policy responses, focusing on those currently being explored by policy institutions. We find that neither an interoperability mandate nor public cost-sharing always achieve the socially optimal outcome when implemented in isolation. An interoperability mandate reduces the private return to forming a tokenized market, since the inability to exclude other brokers precludes trade diversion. This disincentive addresses excessive investment. However, when insufficient tokenization is the problem, the disincentive from interoperability can be too strong and instead cause an underinvestment problem, since tokenization generates positive externalities for some investors. Public cost-sharing can stimulate investment by reducing the burden falling on brokers, but—when used alone—it does not affect the trade diversion incentives that produce exclusive coalitions. In combination, an interoperability mandate and public cost-sharing give the policymaker both a "carrot" and a "stick" to achieve the optimal degree of tokenization in all cases.

We highlight two especially promising avenues for future work. First, future work could build on our framework by allowing for choice across multiple tokenized means of payment. Tokenized money is needed to enable trading on a market with tokenized assets (Chiu and Monnet, 2024; Huang and Keister, 2026). Private tokenized monies, including stablecoins, have proliferated, but are also vulnerable to fragmentation (Shin, 2026). Many policy initiatives, in turn, seek to provide a public alternative.[34] Indeed, such efforts could provide an additional source of public support for private tokenization initiatives, similar to our subsidy σ (BIS, 2025; Hebert, Moshammer and Barth, 2023; Maechler and Wehrli, 2021; Patel et al., 2026). Second, future work could allow for richer broker business models, such as monetizing transaction data generated on private tokenized markets, building on existing work on the data economy (e.g., Acemoglu et al., 2022; Agur, Ari and Dell'Ariccia, 2025; Bergemann and Bonatti, 2024; Farboodi and Veldkamp, 2026).

[34] See, for instance, efforts underway to provide tokenized central-bank money for settlement on tokenized markets in Australia, Brazil, Canada, the Eurozone, Hong Kong, India, Indonesia, Japan, Korea, Malaysia, the Philippines, Singapore, South Africa, Switzerland, Thailand, and Uganda. Similarly, Treasury issuance of tokenized bonds has been explored in Hong Kong, the Philippines, Slovenia, and Thailand (Aldasoro et al., 2025; IMF, 2026; Lo, Volz and Haahr, 2026), and is planned in Cambodia, Korea, and the UK.

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ONLINE APPENDIX

A Proofs

This appendix contains proofs omitted from the main text. Full calculations for all steps are performed in a *Mathematica* file that is available on request. That file performs all calculations in exact form, while here we round quantities to two decimal places for conciseness and show the exact (fraction) form only for key results.

A.1 Proof of Lemma 1

A.1.1 Solution approach.

We solve for the equilibrium of the model by backward induction through the ex-ante decision stages (namely, Stages 1-3, since Stage 4 is the type realization stage, while Stage 5 decisions are shown in Table 3 in Section 2.3). As discussed in Section 2.4.2, investors act from the prior that all other investors choose their local brokers, which we will call "staying" (i.e., on-island). If, based on such a status-quo prior, a unique fulfilled-expectations Subgame-Perfect Nash Equilibrium (SPNE) is found then, regardless of whether other equilibria can exist, the selection criterion in Section 2.4.2 implies that this equilibrium is selected. We will refer to this equilibrium as the "Stay Equilibrium". Thus, our solution technique uses the following approach:

1. At Stage 3, under the status-quo prior about other investors' behavior, each investor chooses a broker based on observed broker fees and expected match probabilities.

2. At Stage 2, each broker determines its fees, internalizing how these affect investor sorting at Stage 3. At this stage, brokers take the coalition structure as given.

3. At Stage 1, brokers negotiate about coalition formation, foreseeing the impact of each coalition structure on fees and investor sorting, and the implications thereof for expected profits. We compare the expected profits $\pi_j^e(\mathcal{C}, s, \gamma, \kappa)$ of each broker across coalition structures to identify the coalition structure that results in Stage 1, for each combination of s, γ and κ.

4. If this backward induction results in a unique SPNE that, moreover, is consistent with the status-quo prior (i.e., in which at Stage 3 each investor indeed opts to stay at her local broker), then our derivations end.

Since we will find below that a fulfilled-expectations SPNE based on the status-quo prior always exists, no additional steps (identifying how to proceed in the absence of such a SPNE) to the above algorithm need to be specified.

A.1.2 Detailed proof.

Stage 3. In this stage, each investor li chooses a broker j based on brokers' fees $\boldsymbol{f}$ and the investor's expected match probabilities P^e_{aj} and P^e_{bj} in the event that they join j and are assigned each type:

$$u^e_{li}(\mathcal{C}, \boldsymbol{f}) = \eta + \max\{(1 - f_{lj})\left(\tfrac{1}{2}P^e_{aj} + \tfrac{1}{2}P^e_{bj}\right), 0\} - \tau_{lj}\ . \tag{A.1.1}$$

Since each investor is equally likely to be assigned type a as type b, the expected overall distribution of types (e.g., six a, five a and one b, four a and two b, ...) is symmetric in a and b, so we can write $P^e_j \coloneqq P^e_{aj} = P^e_{bj}$. The expected match probability for an investor choosing broker j depends on two things: the (observed) coalition structure $\mathcal{C}$ and the distribution of investors across brokers. We denote the latter by $\boldsymbol{\Phi}$, defined as a vector whose six elements correspond to the brokers chosen by each investor li, ordered by l and then i. For example, if $\boldsymbol{\Phi}' = (1, 2, 2, 3, 3, 3)$ then we have that each investor stays with their local broker. For convenience below, we denote this particular example the "Stay Distribution", $\boldsymbol{\Phi}^{Stay}$. Since investors choose brokers simultaneously, each investor makes their decisions on the basis of their *expectations* of other investors' broker choices. Thus we further decompose $\boldsymbol{\Phi}$ into two elements: the broker choice $j_{li} \in \{1, 2, 3\}$ of the investor li whose choice we are considering, and that investor's expectations $\boldsymbol{\Phi}^e_{li}$ of the choices of all the other investors. As discussed in our solution technique, we initially confine our attention to cases in which each investor expects all other investors to stay with their local broker, which we denote by $\boldsymbol{\Phi}^e_{li} = \boldsymbol{\Phi}^{Stay}_{li}$. For example, $\boldsymbol{\Phi}^{Stay}_{11} = (j_{11}, 2, 2, 3, 3, 3)$, where j_{11} is the broker choice of investor 11, while that investor acts from the prior that the other investors stay at their local brokers. Moreover, since investors within a given broker l are identical before (Stage 4) type assignment, the expected payoff for each is identical, allowing us to drop the subscript i. Equation (A.1.1) becomes:

$$u^e_{l}(\mathcal{C}, \boldsymbol{f}) = \eta + \max\{(1 - f_{lj})P^e_j(\mathcal{C}, j_l, \boldsymbol{\Phi}^{Stay}_l), 0\} - \tau_{lj}\ . \tag{A.1.2}$$

We first derive the match probabilities $P^e_j(\mathcal{C}, j_l, \boldsymbol{\Phi}^{Stay}_l)$. These can be calculated mechanically in three steps. First, we calculate the relative probability of different numbers of investors being assigned to type a versus b at Stage 4. Since each investor can draw one of two types and investors' draws are independent, there are $2^6 = 64$ possible outcomes. The distribution of the number of type a investors follows from a straightforward application of the binomial distribution and is shown in Table A.1. Second, conditioning on these possible

Table A.1: Odds of investor type distributions

Distribution	Probability
$6a$	1/64
$5a+b$	6/64
$4a+2b$	15/64
$3a+3b$	20/64
$2a+4b$	15/64
$a+5b$	6/64
$6b$	1/64

aggregate outcomes, we derive the match probability of an investor from a given island in each of three scenarios: the "Stay" scenario ($j_l = l$) where an investor chooses her local broker at Stage 3, and the two alternative "Move" scenarios ($j_l \neq l$) where an investor chooses one of the off-island brokers at Stage 3. Finally, we weigh these second-step results by the probabilities of each first-step outcome to derive nine overall match probabilities, one for each island & chosen-broker pair lj.

We repeat this process for each of the five possible coalition structures $\mathcal{C} \in \mathcal{P}$. The resulting probabilities $P^e_j(\mathcal{C}, j_l, \mathbf{\Phi}^{Stay}_l)$ are shown in Table A.2. With these probabilities in hand, we can derive the "Stay Conditions" that must be satisfied if all investors are to remain at their local brokers. Equation (A.1.2) implies that each investor on island l will choose her local broker in Stage 3 if the investor's utility from doing so is no larger than the investor's maximum utility from moving to a new broker j:

$$\eta + \max\{(1 - f_{ll})P^e_l(\mathcal{C}, l, \mathbf{\Phi}^{Stay}_l), 0\} \geq \eta + \max_{j \neq l}\{(1 - f_{lj})P^e_j(\mathcal{C}, j, \mathbf{\Phi}^{Stay}_l), 0\} - \tau_{lj} \quad \text{(A.1.3)}$$

These conditions are shown explicitly by coalition structure in Table A.3.

Stage 2. Brokers set their fees $\boldsymbol{f}_j$ to maximize $\pi^e_j(\mathcal{C}, \boldsymbol{f})$:

$$\max_{\boldsymbol{f}_j}\{\boldsymbol{f}'_j\mathbf{n}^e_j(\mathcal{C}, \boldsymbol{f}) - (\gamma + \mathcal{T} \cdot \kappa) \cdot n^e_{j,L}(\mathcal{C}, \boldsymbol{f}) - s_j\} \,. \quad \text{(A.1.4)}$$

Table A.2: Investor match probabilities

Coalition structure $\mathcal{C}$	Match probability for l investor choosing broker j			
	$l \setminus j$	1	2	3
$\{\{1\},\{2\},\{3\}\}$	1	0.55	0.69	0.70
	2	0.73	0.73	0.76
	3	0.69	0.71	0.71
	$l \setminus j$	1	2	3
$\{\{1,2\},\{3\}\}$	1	0.56	0.69	0.70
	2	0.75	0.75	0.70
	3	0.70	0.73	0.69
	$l \setminus j$	1	2	3
$\{\{1,3\},\{2\}\}$	1	0.61	0.69	0.70
	2	0.74	0.66	0.77
	3	0.70	0.69	0.73
	$l \setminus j$	1	2	3
$\{\{1\},\{2,3\}\}$	1	0.50	0.69	0.70
	2	0.66	0.74	0.77
	3	0.66	0.71	0.71
	$l \setminus j$	1	2	3
$\{\{1,2,3\}\}$	1	0.55	0.69	0.70
	2	0.73	0.73	0.76
	3	0.69	0.71	0.71

Table A.3: Stay Conditions by coalition structure

Coalition structure $\mathcal{C}$	Stay Conditions
$\{\{1\},\{2\},\{3\}\}$	$(1-f_{11})\cdot 0.55 \geq \max\{(1-f_{12})\cdot 0.69,(1-f_{13})\cdot 0.70,0\}-\tau$ $(1-f_{22})\cdot 0.73 \geq \max\{(1-f_{21})\cdot 0.73,(1-f_{23})\cdot 0.76,0\}-\tau$ $(1-f_{33})\cdot 0.71 \geq \max\{(1-f_{31})\cdot 0.69,(1-f_{32})\cdot 0.71,0\}-\tau$
$\{\{1,2\},\{3\}\}$	$(1-f_{11})\cdot 0.56 \geq \max\{(1-f_{12})\cdot 0.69,(1-f_{13})\cdot 0.70,0\}-\tau$ $(1-f_{22})\cdot 0.75 \geq \max\{(1-f_{21})\cdot 0.75,(1-f_{23})\cdot 0.70,0\}-\tau$ $(1-f_{33})\cdot 0.69 \geq \max\{(1-f_{31})\cdot 0.70,(1-f_{32})\cdot 0.73,0\}-\tau$
$\{\{1,3\},\{2\}\}$	$(1-f_{11})\cdot 0.61 \geq \max\{(1-f_{12})\cdot 0.69,(1-f_{13})\cdot 0.70,0\}-\tau$ $(1-f_{22})\cdot 0.66 \geq \max\{(1-f_{21})\cdot 0.74,(1-f_{23})\cdot 0.77,0\}-\tau$ $(1-f_{33})\cdot 0.73 \geq \max\{(1-f_{31})\cdot 0.70,(1-f_{32})\cdot 0.69,0\}-\tau$
$\{\{1\},\{2,3\}\}$	$(1-f_{11})\cdot 0.50 \geq \max\{(1-f_{12})\cdot 0.69,(1-f_{13})\cdot 0.70,0\}-\tau$ $(1-f_{22})\cdot 0.74 \geq \max\{(1-f_{21})\cdot 0.66,(1-f_{23})\cdot 0.77,0\}-\tau$ $(1-f_{33})\cdot 0.71 \geq \max\{(1-f_{31})\cdot 0.66,(1-f_{32})\cdot 0.71,0\}-\tau$
$\{\{1,2,3\}\}$	$(1-f_{11})\cdot 0.55 \geq \max\{(1-f_{12})\cdot 0.69,(1-f_{13})\cdot 0.70,0\}-\tau$ $(1-f_{22})\cdot 0.73 \geq \max\{(1-f_{21})\cdot 0.73,(1-f_{23})\cdot 0.76,0\}-\tau$ $(1-f_{33})\cdot 0.71 \geq \max\{(1-f_{31})\cdot 0.69,(1-f_{32})\cdot 0.71,0\}-\tau$

Since any tokenized market setup cost s_j is sunk by Stage 2, each broker's problem is to balance revenue per trade (which rises with the fees they set) against the number of trades that they expect to facilitate (which falls when fee increases induce investor movements away from the broker). For each coalition structure, we proceed in two steps. First, we derive the "best competing alternative" that is offered to an investor at l by one of their non-local brokers $j \neq l$. In most cases this (Bertrand competition outcome) turns out to be $f_{lj} = 0$, reflecting that each broker is generally willing to acquire additional investors even at a zero fee because doing so increases the match probability of the broker's local investors, which in turn raises the fee the broker can charge them. Second, we note that this best competing alternative defines the reservation utility that each broker must offer its local investors in order to retain them. The best competing alternative thus pins down the right-hand side of each Stay Condition in Table A.3. Since brokers always charge the highest fee consistent with retaining their local investors, they set fees such that each Stay Condition binds with equality, allowing us to rearrange each condition to derive the optimal "stayer" fee f_{ll} in each case.

No Coalition ($\{\{1\}, \{2\}, \{3\}\}$). First consider the fees that Brokers 1 and 2 charge their own local investors—i.e., f_{11} and f_{22}. From the first two Stay Conditions we see that Broker 3 is the most attractive alternative broker for those investors when considering match probabilities alone. To ascertain whether Broker 3 would bid down to $f_{13} = 0$ and $f_{23} = 0$, consider the following three cases:

(i) Both Brokers 1 and 2 charge the f_{11} and f_{22} that result from the Stay Conditions when $f_{13} = 0$ and $f_{23} = 0$;

(ii) Both Brokers 1 and 2 charge fees above the level in case (i);

(iii) Only one of Broker 1 and 2, denoted as Broker X, charges fees above the level in case (i), while the other, denoted Broker Y, charges fees at the level in case (i).

In case (ii), Brokers 1 and 2 are charging fees above the level implied by the Stay Conditions, meaning Broker 3 could attract both brokers' local investors by setting $f_{13} = 0$ and $f_{23} = 0$—and doing so would be profitable for Broker 3, since attracting all investors implies clearing all trades intra-broker at zero marginal cost. Thus Brokers 1 and 2 do not charge the case (ii) fees in equilibrium. In case (iii), Broker 3 cannot undercut Broker X by charging $f_{X3} = 0$, since Broker Y maintains their local investors and so Broker 3 could incur positive marginal costs when facilitating remaining legacy market transactions. Thus there exists a level of excess fees—i.e., fees above those in case (i)—that Broker X can charge at which it will not be undercut by Broker 3. This in turn implies that Broker 3's threat to charge $f_{Y3} = 0$ is not credible, since Broker 3 could incur positive marginal costs when facilitating legacy market transactions with Broker X. Thus Broker Y can also charge fees above the level in case (i), so case (iii) collapses into case (ii) and the only stable equilibrium is that which arises in case (i)—i.e., $f_{13} = 0$, $f_{23} = 0$ and

$$f_{11} = 1.83\,\tau - 0.29 \tag{A.1.5}$$

$$f_{22} = 1.38\,\tau - 0.04\ . \tag{A.1.6}$$

Lastly, we turn to f_{33} and consider the "best competing alternative" to Broker 3 that would be offered by Brokers 1 and 2. It can be shown that: (i) Broker 1 is willing to set $f_{31} = 0$, (ii) Broker 2 is not willing to set $f_{32} = 0$, but (iii) even so, Broker 2 is willing to set f_{32} sufficiently low that—when combined with the higher match probability that it offers relative to Broker 1—Broker 2 remains the best competing alternative. Specifically, Broker 2 is willing to lower its fee until $f_{32} = 0.02\gamma$, which when combined with the Stay Conditions for the No Coalition case implies that Broker 3 sets

$$f_{33} = 1.41\,\tau + 0.02\,\gamma\ . \tag{A.1.7}$$

1 & 2 Coalition ($\{\{1, 2\}, \{3\}\}$). If either Broker 1 or Broker 2 can attract Broker 3's local investors, their marginal cost of processing transactions will always be zero since there will be no more legacy trades. Thus Brokers 1 and 2 are willing to bid down to $f_{31} = f_{32} = 0$, which when combined with Broker 3's Stay Condition gives

$$f_{33} = 1.45\,\tau - 0.07\ . \tag{A.1.8}$$

Next, it can be shown that the number of legacy market trades that Broker 3 expects to clear *falls* if it can attract Broker 1's local investor, since Broker 3's greater likelihood of intra-broker matches outweighs the effect of hosting more investors who could require an inter-broker match. Thus Broker 3 always gains from acquiring Broker 1's investor, even if it earns zero fees from them directly, since the acquisition increases Broker 3's expected number of fee-paying intra-broker matches *and* lowers Broker 3's expenses. Thus, Broker 3 competes down to $f_{13} = 0$, which combines with Broker 1's Stay Condition to give

$$f_{11} = 1.78\,\tau - 0.25\ . \tag{A.1.9}$$

Lastly, we turn to Broker 2's local investors. The second Stay Condition reveals that Broker 1 is the most competitive off-island broker for these investors. It can further be shown that

Broker 1's profits are higher in the "Move" case in which it sets $f_{21} = 0$ and attracts Broker 2's investors than in the "Stay" case in which it does not attract Broker 2's investors. Thus, Broker 1 and Broker 2 compete on fees down to the point where Broker 1 sets $f_{21} = 0$, which by the second Stay Condition implies that Broker 2 sets

$$f_{22} = 1.33\,\tau\ . \tag{A.1.10}$$

1 & 3 Coalition ($\{\{1,3\},\{2\}\}$). First, an analogous argument to that from cases (i)-(iii) in the No Coalition scenario implies that $f_{13} = 0$ and $f_{23} = 0$ in equilibrium, and thus from the Stay Conditions we have:

$$f_{11} = 1.64\,\tau - 0.15 \tag{A.1.11}$$

$$f_{22} = 1.52\,\tau - 0.17\ . \tag{A.1.12}$$

Turning to f_{33}, we again consider the "best competing alternative" to remaining at Broker 3 for Broker 3's local investors. From the Stay Conditions, we see that their match probability is higher at Broker 1 than at Broker 2. Calculating Broker 1's profits under the scenarios that (i) Broker 3's local investors stay at Broker 3 and (ii) Broker 1 attracts Broker 3's local investors by setting $f_{31} = 0$, we find that Broker 1 prefers the latter. Thus Broker 1 is willing to compete with Broker 3 on fees down to $f_{31} = 0$, so Broker 3's fees are determined by the corresponding Stay Condition, giving:

$$f_{33} = 1.36\,\tau + 0.04\ . \tag{A.1.13}$$

2 & 3 Coalition ($\{\{2,3\},\{1\}\}$). Again, an analogous argument to that from cases (i)-(iii) in the No Coalition scenario implies that $f_{13} = 0$ and $f_{23} = 0$ in equilibrium, and thus from the Stay Conditions we have:

$$f_{11} = 2.00\,\tau - 0.41 \tag{A.1.14}$$

$$f_{22} = 1.35\,\tau - 0.04\ . \tag{A.1.15}$$

Turning to f_{33}, from the match probabilities we see that Broker 2 is again the best off-island alternative for Broker 3's local investors. Furthermore, it can be shown that Broker 2 is willing to compete down to $f_{32} = 0$, implying a Bertrand fee setting game with Broker 3—which in turn implies that Broker 3's equilibrium fees are given by:

$$f_{33} = 1.40\,\tau\ . \tag{A.1.16}$$

Grand Coalition ($\{\{1, 2, 3\}\}$). Since all brokers are part of the tokenized platform, the legacy technology is never used and the marginal cost of processing transactions is always zero. Thus non-local brokers are always willing to bid down to $f_{lj} = 0$ (for all $l \neq j$), and a Bertrand race between them and the local broker l ensures that this occurs in equilibrium. Thus the Stay Conditions become:

$$(1 - f_{11}) \cdot 0.55 = \max\{0.69, 0.70, 0\} - \tau \tag{A.1.17}$$

$$(1 - f_{22}) \cdot 0.73 = \max\{0.73, 0.76, 0\} - \tau \tag{A.1.18}$$

$$(1 - f_{33}) \cdot 0.71 = \max\{0.69, 0.71, 0\} - \tau \tag{A.1.19}$$

which rearrange to give

$$f_{11} = 1.83\,\tau - 0.29 \tag{A.1.20}$$

$$f_{22} = 1.38\,\tau - 0.04 \tag{A.1.21}$$

$$f_{33} = 1.41\,\tau\ . \tag{A.1.22}$$

Summary. We have shown that for each coalition structure there exists a unique set of fees charged such that both (i) all brokers set fees to maximize expected profits and (ii) all investors maximize expected utility by choosing to stay, under the prior that all other investors will stay. Thus, for any coalition structure $\mathcal{C}$ that emerges from Stage 1, there is a unique equilibrium of the resulting subgame. Since all brokers foresee this equilibrium, and their corresponding expected profits $\pi_j^e(\mathcal{C})$, these are the payoffs that they consider when negotiating in Stage 1—to which we now turn.

Stage 1. We first calculate each broker's expected profits from each coalition structure, shown in Table A.4.

We proceed through the iterated deletion of dominated coalition structures. First, we consider the 1 & 2 Coalition. Comparing the second and third rows of Table A.4 reveals that Broker 1 always prefers the 1 & 3 Coalition to the 1 & 2 Coalition, for any values of the parameters. Broker 3 also prefers the 1 & 3 Coalition unless:

$$\pi_3^e(1\ \&\ 3) = 3\tau - 0.20(\gamma + \kappa) - 0.50\,s + 0.09 < \pi_3^e(1\ \&\ 2) = 3\tau - 0.56(\gamma + \kappa) - 0.14$$

$$\iff s > 0.72(\gamma + \kappa) + 0.46\ . \tag{A.1.23}$$

However, if this condition does hold, then we can also see (by substituting it into $\pi_1^e(1\ \&\ 2)$ in Table A.4) that Broker 1 would not choose to form the 1 & 2 Coalition in the first place,

Table A.4: Brokers' expected profits by coalition structure

Coalition structure $\mathcal{C}$	$\pi_1^e(\mathcal{C})$	$\pi_2^e(\mathcal{C})$	$\pi_3^e(\mathcal{C})$
$\{\{1\},\{2\},\{3\}\}$	$\tau - 0.55\gamma - 0.16$	$2\tau - 0.45\gamma - 0.06$	$3\tau - 0.58\gamma$
$\{\{1,2\},\{3\}\}$	$\tau - 0.31(\gamma+\kappa) - 0.50\,s - 0.14$	$2\tau - 0.25(\gamma+\kappa) - 0.50\,s$	$3\tau - 0.56(\gamma+\kappa) - 0.14$
$\{\{1,3\},\{2\}\}$	$\tau - 0.11(\gamma+\kappa) - 0.50\,s - 0.09$	$2\tau - 0.31(\gamma+\kappa) - 0.23$	$3\tau - 0.20(\gamma+\kappa) - 0.50\,s + 0.09$
$\{\{1\},\{2,3\}\}$	$\tau - 0.50(\gamma+\kappa) - 0.20$	$2\tau - 0.17(\gamma+\kappa) - 0.50\,s - 0.06$	$3\tau - 0.33(\gamma+\kappa) - 0.50\,s$
$\{\{1,2,3\}\}$	$\tau - 0.33\,s - 0.16$	$2\tau - 0.33\,s - 0.06$	$3\tau - 0.33\,s$

since under that condition it strictly prefers No Coalition. Thus the 1 & 2 Coalition never arises in equilibrium: either Brokers 1 and 3 prefer to instead form the 1 & 3 Coalition, or Broker 1 prefers to not form any coalition.

Second, we consider the 2 & 3 Coalition. Comparing the third and fourth rows of Table A.4 reveals that Broker 3 always prefers the 1 & 3 Coalition to the 2 & 3 Coalition. Broker 1 also prefers the 1 & 3 Coalition unless:

$$\pi_1^e(1\ \&\ 3) = \tau - 0.11(\gamma+\kappa) - 0.50\,s - 0.09 < \pi_1^e(2\ \&\ 3) = \tau - 0.50(\gamma+\kappa) - 0.20$$
$$\iff s > 0.78(\gamma+\kappa) + 0.22\ . \tag{A.1.24}$$

However, if this condition does hold, then we can see (by substituting it into $\pi_3^e(2\ \&\ 3)$ in Table A.4) that Broker 3 would not choose to form the 2 & 3 Coalition in the first place, since under that condition they strictly prefer No Coalition. Thus the 2 & 3 Coalition also never arises in equilibrium: either Brokers 1 and 3 prefer to instead form the 1 & 3 Coalition, or Broker 3 prefers to not form any coalition.

Next we turn to the Grand Coalition. The unique attraction of this outcome for the brokers is its elimination of the legacy transaction cost γ (eliminating κ is also an attraction relative to the partial coalitions but not compared to No Coalition). Thus, the Grand Coalition is most attractive relative to the remaining coalition structures when this cost is maximized, i.e., when $\gamma = \frac{1}{6}$.[35] In this case, Brokers 1 and 3 would nonetheless prefer to

[35] And by (7) this then implies $\kappa \to 0$. Note that the combination of $\gamma = \frac{1}{6}$ and $\kappa \to 0$ is sufficient here because it maximizes the benefit of the Grand Coalition relative to both partial coalitions (the gap to which is largest for $\gamma + \kappa = \frac{1}{6}$ and this is attained by $\gamma = \frac{1}{6}$ and $\kappa \to 0$) and No Coalition.

deviate to form the 1 & 3 Coalition unless:

$$\pi_1^e(1\ \&\ 3) = \tau - 0.11 * \left(\frac{1}{6}\right) - 0.50\,s - 0.09 < \pi_1^e(\text{Grand}) = \tau - 0.33\,s - 0.16$$
$$\iff s > 0.30 \tag{A.1.25}$$

i.e., setup costs are so high that Broker 1 prefers to share them through the Grand Coalition.[36] However, even in this case the Grand Coalition would not form, since when $\gamma = \frac{1}{6}$ we have $\pi_1^e(\text{No Coalition}) = \tau - 0.25$ yet when $s > 0.30$ we have $\pi_1^e(\text{Grand}) < \tau - 0.26$. Thus at least one Broker always prefers to deviate from the Grand Coalition, so it never forms in equilibrium. Either the cost s of forming a tokenized market is low enough that Brokers 1 and 3 prefer to form their own partial coalition, or s is so high that at least one broker prefers not to form any coalition.

Having established that only No Coalition and the 1 & 3 Coalition can form in equilibrium, we identify the parameter ranges for which each occurs. Comparing the first and third rows of the Table A.4, Broker 3 prefers the 1 & 3 Coalition over No Coalition when

$$\pi_3^e(1\ \&\ 3) = 3\tau - 0.20(\gamma + \kappa) - 0.50\,s + 0.09 > \pi_3^e(\text{No Coalition}) = 3\tau - 0.58\gamma$$
$$\iff s < 0.18 + 0.76\gamma - 0.4\kappa \tag{A.1.26}$$

and that Broker 1 prefers the 1 & 3 Coalition over No Coalition when

$$\pi_1^e(1\ \&\ 3) = \tau - 0.11(\gamma + \kappa) - 0.50\,s - 0.09 > \pi_1^e(\text{No Coalition}) = \tau - 0.55\gamma - 0.16$$
$$\iff s < 0.14 + 0.88\gamma - 0.22\kappa\ , \tag{A.1.27}$$

The identity of the marginal broker depends on which of the two terms on the right-hand side dominates, i.e., which broker has the tighter— and we find that the condition is always tighter for Broker 1.[37] Expressing Broker 1's condition in exact fractions:

$$s < \tfrac{1}{8} + \tfrac{7}{8}\gamma - \tfrac{7}{32}\kappa\ . \tag{A.1.28}$$

When s is below $\frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$ the 1 & 3 Coalition forms, while when s is above that level No

[36] We note that whenever s is sufficiently low that Broker 1 prefers to form the 1 & 3 Coalition, so does Broker 3, since Broker 3 prefers the 1 & 3 Coalition whenever $\pi_3^e(1\ \&\ 3) > \pi_3^e(\text{Grand})$, which for $\gamma = \frac{1}{6}$ yields $s < 0.33$, i.e., Broker 1 is the marginal broker.

[37] Within the allowed parameter space, the constraint is tightest on Broker 3 relative to Broker 1 when $\gamma = \kappa = \frac{1}{12}$ at which point Broker 3's threshold level of s is 0.21, whereas Broker 1's threshold level of s is 0.20. Thus, Broker 1's threshold is always below (i.e., tighter than) that of Broker 3, so Broker 1's condition (i.e., equation A.1.28) forms the boundary.

Coalition forms. When $s = \frac{1}{8} + \frac{7}{8}\gamma - \frac{7}{32}\kappa$, Broker 1 is indifferent and so the 1 & 3 Coalition forms as described on page 15.

A.2 Proof of Lemma 2

Expected aggregate welfare $W^e(\mathcal{C})$ for each coalition structure is the sum of four terms. First is the sum of investors' endowments 6η: these are either consumed or paid to investors in fees. Second is the total expected gains from trade. This can be calculated as 4.125, which is the maximum number of matches that can take place for each distribution in Table A.1, multiplied by the probability of that distribution occurring. This number is constant across coalition structures, since the coalition structure only affects the location of the matches, not their total quantity. The third term is the cost of creating a tokenized market, s, which is incurred under all coalition structures except No Coalition. The final term is the expected total cost $(\gamma + \mathcal{T} \cdot \kappa) \cdot n_L^e$ of processing legacy market trades, where $n_L^e = n_{1,L}^e + n_{2,L}^e + n_{3,L}^e$ is the total number of such trades expected to be facilitated across all three brokers.[38]

Table A.5 summarizes the resulting welfare expressions for each coalition structure. It is immediately apparent that no partial coalition structure ever maximizes welfare. It can be shown that $n_L^e(\text{No Coalition}) = \frac{13}{8}$, so the set of points for which welfare is equal under the No Coalition and the Grand Coalition is given by:

$$\begin{aligned} W^e(\text{No Coalition}) = 6\eta + 4.125 - \gamma \cdot n_L^e(\text{No Coalition}) = W^e(\text{Grand}) = 6\eta + 4.125 - s \\ \iff s = \frac{13}{8}\gamma\,. \end{aligned} \tag{A.2.1}$$

Thus, when $s > \frac{13}{8}\gamma$ welfare is maximized by No Coalition, when $s < \frac{13}{8}\gamma$ welfare is maximized by the Grand Coalition, and welfare is equal under both coalition structures when $s = \frac{13}{8}\gamma$.

A.3 Proof of Lemma 3

The proof proceeds in two stages. First we solve the game when the interoperability mandate is imposed, then we relate the results to the outcomes of the baseline model.

A.3.1 Equilibrium under the interoperability mandate

Stages 3 and 2 of the game are identical to those described in the proof of Lemma 1. Thus, brokers' payoffs from coalition structure $\mathcal{C}$, if it emerges from the new (three-sub-stage) Stage

[38] Note τ does not appear in expected aggregate welfare because it is never incurred in the Stay Equilibrium.

Table A.5: Welfare by coalition structure

Coalition structure $\mathcal{C}$	Expected aggregate welfare $W^e(\mathcal{C})$
$\{\{1\},\{2\},\{3\}\}$	$6\eta + 4.125 - \gamma \cdot n_L^e(\text{No Coalition})$
$\{\{1,2\},\{3\}\}$	$6\eta + 4.125 - s - (\gamma+\kappa) \cdot n_L^e(1 \text{ \& } 2)$
$\{\{1,3\},\{2\}\}$	$6\eta + 4.125 - s - (\gamma+\kappa) \cdot n_L^e(1 \text{ \& } 3)$
$\{\{1\},\{2,3\}\}$	$6\eta + 4.125 - s - (\gamma+\kappa) \cdot n_L^e(2 \text{ \& } 3)$
$\{\{1,2,3\}\}$	$6\eta + 4.125 - s$

1, are the same as those given in Table A.4. We thus proceed by backward induction through the Stage 1 sub-stages.

Stage 1.3. This stage only exists if a partial coalition formed at Stage 1.2. We consider each of the three possible cases.

1 & 2 Coalition. If the 1 & 2 Coalition formed at Stage 1.2, Broker 3 faces the following two options. If it does not join the tokenized platform its expected profits are

$$\pi_3^e(1 \text{ \& } 2 \text{ Coalition}) = 3\tau - 0.56(\gamma+\kappa) - 0.14 \tag{A.3.1}$$

as in Table A.4. If instead it chooses to accept the offer to join the coalition at no cost, its profit is the Grand Coalition profit in Table A.4, minus any share of the setup cost—giving $\pi_3^e(\text{Grand via } 1 \text{ \& } 2) = 3\tau$. This dominates $\pi_3^e(1 \text{ \& } 2 \text{ Coalition})$, so Broker 3 chooses to join.

1 & 3 Coalition. By analogous reasoning, in this case Broker 2 faces a choice between

$$\pi_2^e(1 \text{ \& } 3 \text{ Coalition}) = 2\tau - 0.31(\gamma+\kappa) - 0.23 \tag{A.3.2}$$

and $\pi_2^e(\text{Grand via } 1 \text{ \& } 3) = 2\tau - 0.06$. Again, the latter dominates so Broker 2 chooses to join the coalition.

2 & 3 Coalition. Finally, by the same logic Broker 1 chooses between

$$\pi_1^e(\text{2 \& 3 Coalition}) = \tau - 0.50(\gamma + \kappa) - 0.20 \tag{A.3.3}$$

and $\pi_1^e(\text{Grand via 2 \& 3}) = \tau - 0.16$. Again, the latter dominates so Broker 1 chooses to join.

Stage 1.2. Foreseeing that any excluded broker will subsequently accept the offer to join, brokers consider the resulting expected profit structure shown in Table A.6. The first and fifth rows are unchanged relative to the baseline case without interoperability (Table A.4), while the middle rows reflect that partial coalitions all ultimately result in the Grand Coalition forming, simply with different distributions of the setup cost s. To proceed, we now consider the coalition structures that result from negotiation at Stage 1.2 for various different investment budget limits s_j^{max} set by the brokers at Stage 1.

Table A.6: Brokers' expected profits under interoperability mandate

Coalition structure at Stage 1.2	$\pi_1^e(\mathcal{C})$	$\pi_2^e(\mathcal{C})$	$\pi_3^e(\mathcal{C})$
$\{\{1\},\{2\},\{3\}\}$	$\tau - 0.55\gamma - 0.16$	$2\tau - 0.45\gamma - 0.06$	$3\tau - 0.58\gamma$
$\{\{1,2\},\{3\}\}$	$\tau - 0.50\,s - 0.16$	$2\tau - 0.50\,s - 0.06$	3τ
$\{\{1,3\},\{2\}\}$	$\tau - 0.50\,s - 0.16$	$2\tau - 0.06$	$3\tau - 0.50\,s$
$\{\{1\},\{2,3\}\}$	$\tau - 0.16$	$2\tau - 0.50\,s - 0.06$	$3\tau - 0.50\,s$
$\{\{1,2,3\}\}$	$\tau - 0.33\,s - 0.16$	$2\tau - 0.33\,s - 0.06$	$3\tau - 0.33\,s$

All brokers unconstrained. Let $s_1^{max} \geq 0.5$, $s_1^{max} \geq 0.5$ and $s_1^{max} \geq 0.5$. In this case, no broker is constrained in the amount that it can invest in setup costs, so all rows in Table A.6 are feasible. To see the implications, label the three brokers X, Y and Z (in any order) and consider the perspective of Broker X. For any coalition structure not containing the coalition $\{X\}$, they will prefer the alternative coalition structure $\{\{X\},\{Y,Z\}\}$ in which the other brokers form a partial coalition, which X then joins at Stage 1.3. Thus the only possible equilibria are the coalition structures that contain $\{X\}$—i.e., $\{\{X\},\{Y,Z\}\}$ and $\{\{X\},\{Y\},\{Z\}\}$. But since Y and Z reason in the same way, $\{\{X\},\{Y,Z\}\}$ can never form, so the only possible equilibrium is $\{\{X\},\{Y\},\{Z\}\}$.

Two brokers unconstrained. Consider now the case in which any two brokers, Y and Z, have $s_Y^{max} \geq 0.5$ and $s_Z^{max} \geq 0.5$, while the third broker X has $s_X^{max} < 0.5$. Again, for any coalition structure not containing the coalition $\{X\}$, X will prefer the alternative coalition structure $\{\{X\},\{Y,Z\}\}$ in which the other brokers form a partial coalition, which X then joins at Stage 1.3. For Y and Z, the hope of free-riding in this way is not available, since in each case the other two brokers lack sufficient resources to meet the investment cost s alone. Thus the only possible equilibria are the coalition structures that contain $\{X\}$—i.e., $\{\{X\},\{Y,Z\}\}$ and $\{\{X\},\{Y\},\{Z\}\}$.

One or zero brokers unconstrained. If fewer than two brokers have $s_j^{max} \geq 0.5$, free-riding is impossible for all brokers. Given that investment costs are shared equally among coalition members, there are two possible scenarios: (i) if one or more brokers have $s_j^{max} < 0.33$, only No Coalition is possible; (ii) if instead all three brokers have $s_j^{max} \geq 0.33$, the Grand Coalition is also possible. Specifically, the Grand Coalition will form if:

$$\begin{aligned}\pi_1^e(\text{Grand}) = \tau - 0.33\, s - 0.16 &\geq \pi_1^e(\text{No Coalition}) = \tau - 0.55\gamma - 0.16 \\ \iff s &\leq 1.67\gamma \end{aligned} \tag{A.3.4}$$

and

$$\begin{aligned}\pi_2^e(\text{Grand}) = 2\tau - 0.33\, s - 0.06 &\geq \pi_2^e(\text{No Coalition}) = 2\tau - 0.45\gamma - 0.06 \\ \iff s &\leq 1.36\gamma \\ \text{in fractions: } s &\leq \frac{27}{20}\gamma \end{aligned} \tag{A.3.5}$$

and

$$\begin{aligned}\pi_3^e(\text{Grand}) = 3\tau - 0.33\, s &\geq \pi_3^e(\text{No Coalition}) = 3\tau - 0.58\gamma \\ \iff s &\leq 1.76\gamma\ , \end{aligned} \tag{A.3.6}$$

which reduces to the single condition $s \leq \frac{27}{20}\gamma$. Thus Broker 2 is the marginal broker, for whom setup costs must fall the most before the Grand Coalition is preferable to No Coalition.

Stage 1.1. Anticipating the outcomes above, at Stage 1.1 each Broker X foresees that:

1. If they set $s_X^{max} \geq 0.5$, the outcome after Stage 1.3 will be either No Coalition or a Grand Coalition for which X pays $s_X = 0.5$—i.e., more than an equal share;

2. If they set $s_X^{max} < 0.33$, the outcome after Stage 1.3 will be No Coalition or a Grand Coalition for which X pays $s_x = 0$;

3. If they set $0.33 \leq s_X^{max} < 0.5$, there are three possibilities: No Coalition, a Grand Coalition to which X contributes equally ($s_X = 0.33$), or a Grand Coalition to which X contributes nothing ($s_X = 0$).

Policy 1 is dominated by the other two, so X will never choose it. All brokers reason the same way, which in turn rules out the Grand Coalition outcome under Policy 2—since if no broker sets $s_j^{max} \geq 0.5$, there is never an opportunity to free-ride. Thus Policy 2 always produces No Coalition. Similarly, since no broker sets $s_j^{max} \geq 0.5$, the free-riding Grand Coalition outcome under Policy 3 is also ruled out, so Policy 3 always produces either No Coalition or the Grand Coalition to which all brokers contribute equally (i.e., the "Egalitarian Grand Coalition").

In summary, broker j's choices at Stage 1.1 reduce to a choice between allowing the possibility of the Egalitarian Grand Coalition at Stage 1.2 (which requires setting $0.33 \leq s_X^{max} < 0.5$) and ruling it out (by setting $s_X^{max} < 0.33$). Broker 1 does the former if condition (A.3.4) holds; Broker 2 does likewise if condition (A.3.5) holds; and Broker 3 does likewise if condition (A.3.6) holds. If condition (A.3.5) holds, so must the other two, so the outcome depends only on condition (A.3.5)—i.e., on whether Broker 2 is willing to form the Egalitarian Grand Coalition. Thus under the interoperability mandate, the Egalitarian Grand Coalition forms if $s \leq \frac{27}{20}\gamma$, and No Coalition forms otherwise. This outcome is summarized by the green boundary in Figure 4a.

A.3.2 Relation to the outcomes of the baseline model

The result in Lemma 3 follows immediately from comparing the outcome derived in the previous subsection to the excessive investment region presented in Proposition 1.

A.4 Proof of Lemma 4

The result follows immediately from comparing the outcome derived in Section A.3.1 to the insufficient tokenization region presented in Proposition 1. Graphically, comparing Figures 3b and 4a reveals that the lower part of the insufficient tokenization region in the former changes to the optimal Grand Coalition outcome in the latter. The boundary of this lower part is given by the condition $s \leq \frac{27}{20}\gamma$ derived in Section A.3.1.

A.5 Proof of Lemma 5

The result follows immediately from comparing the outcome derived in Section A.3.1 to the insufficient tokenization region presented in Proposition 1. Graphically, comparing Figures 3b and 4a reveals that, in the upper part of the insufficient tokenization region in the former, No Coalition results in the latter. The upper boundary of this part is given by the condition $s \leq \frac{13}{8}\gamma$ in Proposition 1. The lower boundary of this part is given by the condition $s > \frac{27}{20}\gamma$ derived in Section A.3.1.

A.6 Proof of Lemma 6

Let the policymaker bear σ of the total setup cost s when $\frac{27}{20}\gamma < s \leq \frac{13}{8}\gamma$. Following Section A.3.1 above, Broker 2 then chooses the Grand Coalition over No Coalition if:

$$s \leq \frac{27}{20}\gamma + \sigma \; . \tag{A.6.1}$$

From Lemma 2, welfare is maximized when the corresponding boundary is:

$$s \leq \frac{13}{8}\gamma \; . \tag{A.6.2}$$

These two conditions are thus equivalent when:

$$\begin{aligned} \frac{27}{20}\gamma + \sigma &= \frac{13}{8}\gamma \\ \iff \sigma &= \frac{11}{40}\gamma \; . \end{aligned} \tag{A.6.3}$$

Thus, a subsidy of this level or greater prevents the insufficient tokenization that would otherwise result, by tipping Broker 2 into choosing the Grand Coalition.

A.7 Proof of Proposition 3

Stages 3 and 2 are identical to those in the baseline setup. We derive the equilibrium outcomes at Stage 1 in three steps.

First, we note that when side-payments are possible any equilibrium outcome $(\mathcal{C}, \mathbf{v})$ must maximize expected aggregate broker profits. The proof of this statement is by contradiction. Suppose that an equilibrium outcome $(\mathcal{C}, \mathbf{v})$ exists that does not maximize expected joint broker profits Π^e. By definition, there exists some alternative coalition structure $\mathcal{C}' \neq \mathcal{C}$ such that $(\mathcal{C}', \mathbf{v})$ does maximize expected joint broker profits. Since total expected profits are larger, there must then exist some trio of net transfers $\mathbf{v}'$ such that $\pi_j^e(\mathcal{C}', \mathbf{v}', s, \gamma, \kappa) > \pi_j^e(\mathcal{C}, \mathbf{v}, s, \gamma, \kappa)\ \forall\ j$ (i.e., all brokers are strictly better off). Thus there does exist an alternative pair $(\mathcal{C}', \mathbf{v}')$ containing at least one new coalition $C' \notin \mathcal{C}$ such that all brokers in C' are strictly better off than under $(\mathcal{C}, \mathbf{v})$—and the required side-payments will be acceptable to the payers, since all brokers are better off—so $(\mathcal{C}, \mathbf{v})$ cannot be an equilibrium.

Second, we note that expected aggregate broker profits are maximized by: (i) No Coalition when $s \geq \frac{7313}{4640}\gamma$, and (ii) the Grand Coalition when $s \leq \frac{7313}{4640}\gamma$. To see this, we first note that at Stage 1 brokers' expected aggregate profits by coalition structure are the same as those shown in the final column of Table A.4 above, since total side-payments must sum to zero. That column reveals that aggregate profits under the Grand Coalition are always greater than those under any partial coalition. Comparing Π^e(Grand) to Π^e(No Coalition) then reveals that aggregate broker profits are maximized by the Grand Coalition when $s \leq \frac{7313}{4640}\gamma = 1.58\gamma$ and by No Coalition when $s \geq \frac{7313}{4640}\gamma = 1.58\gamma$.

Combining these two steps and applying the tie-breaking assumption described in Section 2.4 gives that, when brokers can make side-payments, for any equilibrium outcome $(\mathcal{C}, \mathbf{v})$ the coalition structure $\mathcal{C}$ must be: (i) No Coalition when $s > \frac{7313}{4640}\gamma$; and (ii) the Grand Coalition when $s \leq \frac{7313}{4640}\gamma$. The proposition then follows from comparing this result to Lemma 2. When $\frac{7313}{4640}\gamma < s < \frac{13}{8}\gamma$, the Grand Coalition alone is socially optimal but a private equilibrium must feature No Coalition, implying insufficient tokenization and underinvestment.

A.8 Proof of Proposition 4

The result follows from the proof of Proposition 3, and the discussion of a tokenization subsidy in Section 5. A subsidy σ induces brokers at point (γ, s) to make the decision that they would otherwise make at point $(\gamma, s-\sigma)$. The proof of Proposition 3 shows that the fully flexible side-payments equilibrium fails to maximize welfare only when brokers' perceived s is too high, such that an equilibrium features No Coalition when the Grand Coalition is socially optimal. It also shows that when brokers' perceived s is sufficiently low, equilibrium must feature the Grand Coalition. Therefore, a sufficiently large subsidy—lowering brokers' perceived s to the point where equilibrium features the Grand Coalition—can always align equilibrium outcomes with social welfare.

A.9 Proof of Proposition 5

Since this proof follows the same structure and steps as for the baseline model, we here present and discuss the summary tables, based on which the results are derived. The full calculations underlying these tables are in the *Mathematica* file that is available on request.

At Stage 4, there are only two possible investor type distributions: either the three investors are of the same type or two investors are of one type and one of the other. As seen from Table A.7, the former occurs with probability $\frac{1}{4}$ and the latter with probability $\frac{3}{4}$.

Table A.7: Odds of investor type distributions

Distribution	Probability
$3a$	1/8
$2a + 1b$	3/8
$a + 2b$	3/8
$3b$	1/8

Table A.8: Investor match probabilities

Coalition structure	Investor match probability
No Coalition	Local broker: $\frac{1}{2}$; Off-island: $\frac{5}{8}$
Partial coalition—includes local broker	Local broker: $\frac{5}{8}$; Off-island: $\frac{5}{8}$
Partial coalition—excludes local broker	Local broker: $\frac{1}{4}$; Off-island: $\frac{5}{8}$
Grand Coalition	Local broker: $\frac{1}{2}$; Off-island: $\frac{5}{8}$

At Stage 3, investors foresee their type probabilities at Stage 4 and, depending on their island location and brokers' coalition structure, they calculate their probability of being matched at Stage 5, and how this depends on the choice between their local broker and off-island brokers, shown in Table A.8. Under a partial coalition, matching probabilities when selecting the local broker differ between investors whose local broker is, respectively, included or excluded from the coalition.

At Stage 2, for any given coalition structure, brokers optimize fees on investors. Brokers here always compete down to zero fees for any off-island investors willing to choose them. In response, brokers set the fees shown in Table A.9 on their local investors—the highest fees that are consistent with investors becoming on-island clients. Under a partial coalition, the fee that each included broker is able to charge its local investor is higher than that which the excluded broker is able to charge its own local investor, since the included brokers offer a higher match probability.

Table A.9: Brokers' optimal fees on local investors

Coalition structure	Coalition insider	Coalition outsider
No Coalition	N/A	$2\tau - \frac{1}{4}$
Partial Coalition	$\frac{8}{5}\tau$	$4\tau - \frac{3}{2}$
Grand Coalition	$2\tau - \frac{1}{4}$	N/A

Table A.10: Brokers' expected profits

Coalition structure	Coalition insider	Coalition outsider
No Coalition	N/A	$\tau - \frac{\gamma}{2} - \frac{1}{8}$
Partial Coalition	$\tau - \frac{\gamma+\kappa}{8} - \frac{s}{2}$	$\tau - \frac{\gamma+\kappa}{4} - \frac{3}{8}$
Grand Coalition	$\tau - \frac{s}{3} - \frac{1}{8}$	N/A

At Stage 1, brokers compare the expected profits under each coalition structure. These expected profits are shown in Table A.10. Being part of the Grand Coalition is always dominated: when the Grand Coalition is more profitable than No Coalition—when $s \leq \frac{3}{2}\gamma$—then it is necessarily also *less* profitable than being an insider on a partial coalition (given also the constraints on γ and κ from (7)). Hence, the Grand Coalition cannot be the equilibrium coalition structure, since two or more brokers will find a profitable deviation from it. No Coalition is the equilibrium when all brokers find it more profitable than becoming insiders on a partial coalition, namely when $\tau - \frac{\gamma}{2} - \frac{1}{8} > \tau - \frac{\gamma+\kappa}{8} - \frac{s}{2}$ which simplifies to $s > \frac{1-\kappa}{4} + \frac{3}{4}\gamma$. Conversely, if $s \leq \frac{1-\kappa}{4} + \frac{3}{4}\gamma$ then a partial coalition is the equilibrium.

Table A.11 shows overall welfare under the different coalition structures. Welfare is always higher under the Grand Coalition than under a partial coalition, as in the baseline model. The Grand Coalition maximizes welfare when $s < \frac{3}{2}\gamma$, while No Coalition dominates when $s > \frac{3}{2}\gamma$. Since a partial coalition forms in the private equilibrium when $s < \frac{1-\kappa}{4} + \frac{3}{4}\gamma$, this implies insufficient tokenization if $s \leq \frac{3}{2}\gamma$ and excessive investment if $\frac{3}{2}\gamma \leq s \leq \frac{1-\kappa}{4} + \frac{3}{4}\gamma$.[39]

Table A.11: Welfare by coalition structure

Coalition structure	Welfare
No Coalition	$3\eta + \frac{3}{2}(1-\gamma)$
Partial Coalition	$3\eta + \frac{3}{2} - s - \frac{2}{3}(\gamma+\kappa)$
Grand Coalition	$3\eta + \frac{3}{2} - s$

An interoperability mandate always achieves the social optimum. By precluding trade diversion through partial coalitions (as in the baseline model), it leaves brokers to weigh the Grand Coalition against No Coalition. Table A.10 shows that brokers then prefer the Grand Coalition if $\tau - \frac{s}{3} - \frac{1}{8} \geq \tau - \frac{\gamma}{2} - \frac{1}{8}$, which simplifies to $s \leq \frac{3}{2}\gamma$. This is exactly the cutoff below which the Grand Coalition is socially optimal. Above the cutoff, No Coalition forms, which is again socially optimal.

This coincidence of privately and socially optimal thresholds reflects that broker fees on local investors are $2\tau - \frac{1}{4}$ under both the Grand Coalition and No Coalition (Table A.9). Investors therefore do not gain, relative to No Coalition, from brokers' decisions to form the Grand Coalition. Thus, under the interoperability mandate, all the benefits of tokenization are internalized by brokers. Moreover, since all brokers are identical, their preferences for when to form the Grand Coalition coincide. Thus, when the Grand Coalition is chosen over No Coalition it must maximize aggregate broker profit, and—since there are no externalities to investors—also maximize welfare.

[39] The excessive investment region always exists, since (7) implies that $\frac{3}{2}\gamma < \frac{1-\kappa}{4} + \frac{3}{4}\gamma$. Existence of the insufficient tokenization region follows directly from $\gamma > 0$.

PUBLICATIONS

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